

Practice Multiple-Choice Tests

EXAMINEE STATEMENT, CERTIFICATION, AND SIGNATURE

1. Read the following **Statement**: By opening this test booklet, I agree to the terms and conditions set forth in the ACT registration booklet or website for this exam, including the arbitration and dispute remedy provisions. I understand that ACT owns the test questions and responses and affirm that I will not share any test questions or responses with anyone by any form of communication. I understand that assuming anyone else's identity to take this test may violate the law and be subject to legal penalty.
2. Copy the **Certification** shown below (only the text in *italics*) on the lines provided. Write in your normal handwriting.

Certification: *I agree to the Statement above and certify that I am the person whose name appears on this form.*

3. Sign your name as you would any official document and enter today's date.

Your Signature

Today's Date



The **ACT**[®]



Form 1267C

Directions

This booklet contains tests in English, Mathematics, Reading, and Science. These tests measure skills and abilities highly related to high school course work and success in college. *CALCULATORS MAY BE USED ON THE MATHEMATICS TEST ONLY.*

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. *DO NOT USE INK OR A MECHANICAL PENCIL.*

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will NOT be penalized for guessing. *IT IS TO YOUR ADVANTAGE TO ANSWER EVERY QUESTION EVEN IF YOU MUST GUESS.*

You may work on each test **ONLY** when your test supervisor tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may NOT look back to a test on which time has already been called, and you may NOT go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may NOT for any reason fill in or alter ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

**DO NOT OPEN THIS BOOKLET
UNTIL TOLD TO DO SO.**



P.O. BOX 168
IOWA CITY, IA 52243-0168

©2012 by ACT, Inc. All rights reserved.
NOTE: This test material is the confidential property of ACT, Inc., and may not be copied, reproduced, sold, or otherwise transferred without the prior express written permission of ACT, Inc.



ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose “NO CHANGE.” In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Dragonfly

The nature trail is six feet wide and seven miles long. It slithers through the forest like a snake curving, and bending along the banks of the river.

1

The county cleared this path and paved it with packed

2

gravel, so they would have a peaceful place to hike and bike.

3

I ride this trail nearly every day—not on a bike, but on “Luigi.” That’s the nickname I gave my

4

motorized wheelchair. 5 Today, Luigi’s battery

1. A. NO CHANGE
B. snake, curving and bending
C. snake curving and bending,
D. snake, curving, and bending,
2. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. path, paving
G. path and then paved
H. path before paving
J. path paved
3. A. NO CHANGE
B. knowing they
C. that they
D. people
4. F. NO CHANGE
G. day; not on a bike
H. day not on a bike
J. day, not on a bike;
5. If the writer were to delete the preceding sentence, the essay would primarily lose:
A. a reason why the narrator is in the forest.
B. a detail important for understanding the essay.
C. a contrast to the lighthearted tone of the essay.
D. nothing at all; this information is irrelevant to the essay.

is fully charged, I know I can go all the way to the end
 6
 of the trail and back. But I always carry a cell phone on
 me just in case.

Luigi's motor moves slowly as we venture along
 7
 the trail. I can hear the gravel quietly crunching beneath

Luigi's rubber wheels. I hear the songs of cardinals in the
 8
 trees and the clamor of crickets in the grasses. I hear the
 murmur of water slipping over time-smoothed rocks. It is
 9
 September, and some of the trees are starting to blush red
 and orange at their tips. The wind ruffles my hair and
 chills my face as I bounce gently, along in my padded
 10
 chair.

Bicyclists streak past in a blur of color and a cloud of
 11

dust I don't understand their hurry. Luigi can go fast, but I
 12
 like to ride slowly, to see like a hovering dragonfly. I want
 to see everything that has changed, grown, bloomed, or
 died since yesterday. Today I notice that a spider has
 woven a web between some honeysuckle bushes by the
 bridge. I see that the bank of vibrant yellow black-eyed
 Susans by the barbed wire fence is starting to dry and fade
 away. I spend an hour; looking and listening and learning.
 13

6. F. NO CHANGE
 G. charged, because of that,
 H. charged, this means that
 J. charged, so
7. Which choice would most logically and effectively emphasize the positive, friendly attitude the narrator has toward Luigi?
 A. NO CHANGE
 B. travels safely
 C. proceeds carefully
 D. purrs softly
8. F. NO CHANGE
 G. You can hear
 H. One can even hear
 J. While hearing
9. A. NO CHANGE
 B. Due to the fact that it is
 C. It turns into the month of
 D. Because it has turned into
10. F. NO CHANGE
 G. gentle, along
 H. gently along
 J. gentle along,
11. Which choice most effectively leads into the new subject of this paragraph?
 A. NO CHANGE
 B. The sun begins to set
 C. Nature always impresses me
 D. Days can go by quickly
12. F. NO CHANGE
 G. dust, however,
 H. dust.
 J. dust,
13. A. NO CHANGE
 B. hour, looking,
 C. hour looking;
 D. hour looking



And now my ride is finished for today. I leave the trail and come out into the open, manicured park at the trails end. There, my older brother helps me out of my ¹⁴ chair and into his waiting van. He puts Luigi in the back, and I return to the world of pavement, streetlights, and traffic. But in my mind, I am still gliding through the forest. I am like the water, flowing over ancient stones. Inside, I am still a dragonfly.

14. F. NO CHANGE
G. trail's
H. trails'
J. trails's

Question 15 asks about the preceding passage as a whole.

15. Suppose the writer's goal had been to write an essay illustrating the pleasure that people can take in nature. Would this essay accomplish that goal?
- A. Yes, because it focuses on a variety of wildflowers that the narrator enjoys.
B. Yes, because it focuses on the narrator's joy at having access to nature.
C. No, because it describes the world of the city as being more important to the narrator.
D. No, because it focuses primarily on the functioning of the narrator's motorized wheelchair.

PASSAGE II

Beneath the Streets of New York

At 2 p.m., on October 27, 1904; thousands of ¹⁶ New York City residents poured into the streets of Manhattan. Their cheers competed with the blare of ferryboat horns and the whistle of power plants. The city was celebrating an incredible engineering feat; the ¹⁷ completion of the first section of the New York City

Subway. 18

16. F. NO CHANGE
G. 2 p.m. on October 27, 1904, thousands
H. 2 p.m., on October 27, 1904; thousands,
J. 2 p.m. on October 27, 1904, thousands,

17. A. NO CHANGE
B. feat, over
C. feat:
D. feat

18. The writer is concerned about the level of detail in the preceding sentence and is considering deleting the phrase "the first section of" from it. If the writer were to make this deletion, the paragraph would primarily lose information that:
- F. reveals how expansive the New York City Subway would become.
G. clarifies that only part of the subway system had been completed by October 27, 1904.
H. makes clear that by October 27, 1904, construction of the second section of the subway was already underway.
J. provides evidence that New York City residents at this celebration believed the entire subway system was complete.



The original subway line was 9.1 miles long and had twenty-eight stations. [A] The first train took twenty-six minutes to complete the route, which ran from City Hall to West 145th Street in under a half an hour. Tens of thousands of New Yorkers could now avoid traffic jams

by traveling underneath the streets. [B]

As early as 1865, there had been proposals for a New York subway, but that took decades to resolve the many political, financial, and technical challenges. The engineer, William Barclay Parsons accepted responsibility for overseeing this project.

Parsons decided that most of the subway tunnel would be constructed using an innovation engineering method known as “cut and cover.” [C] First, workers used picks and shovels to remove roads and dig a deep trench. After installing wooden braces to hold back the earth, workers built a concrete floor. Tunnel walls were

created: with layers of brick, ceramic blocks, tar-soaked felt for waterproofing, and concrete. The roof was made from arch-shaped wooden molds also covered with concrete. Next, track beds were filled with crushed stone, and rails were secured to wooden ties. Finally, the roof was covered with tar-soaked felt, and the roads were rebuilt.

19. A. NO CHANGE
B. in the completion of its route.
C. in twenty-six minutes.
D. DELETE the underlined portion and end the sentence with a period.
20. Which choice would most effectively conclude the sentence by indicating clearly how the subway system could address the problem described in the first part of the sentence?
F. NO CHANGE
G. traveling more effectively.
H. trying something new.
J. using a system.
21. A. NO CHANGE
B. it
C. those
D. DELETE the underlined portion.
22. F. NO CHANGE
G. engineer—William Barclay Parsons
H. engineer William Barclay Parsons,
J. engineer William Barclay Parsons
23. A. NO CHANGE
B. innovate engineer
C. innovative engineering
D. innovate engineering
24. F. NO CHANGE
G. into the ground deeply under where the roads had previously been removed by them.
H. a trench far down below since it was necessary to shovel deep into the earth in this method known as “cut and cover.”
J. DELETE the underlined portion and end the sentence with a period.
25. A. NO CHANGE
B. created, with
C. created with
D. created with:



Brightly lit stations welcomed the public, many
of them were skeptical of traveling underground. [D] It
²⁶

didn't take long for New Yorkers to adapt, ²⁷however. The
day after the subway opened, one newspaper reported that
the riders were emerging from underground "having
finished what will be to them the daily routine of the
rest of their lives." ²⁸

26. F. NO CHANGE
G. of whom
H. of who
J. DELETE the underlined portion.

27. A. NO CHANGE
B. therefore.
C. for instance.
D. that is.

28. The writer wishes to add a sentence that describes the magnitude and expansiveness of the New York City Subway system today. Given that all the following statements are true, which one, if added here, would most clearly and effectively accomplish the writer's goal?
- F. Even today, for many New Yorkers that newspaper's account is right!
- G. Today, riding a portion of the New York City Subway's 656 miles of mainline track is a daily routine for more than 4 million people.
- H. Today, the New York City Transit Authority continuously maintains two separate fleets of subway cars.
- J. Now, a typical New York City Subway waiting platform ranges from 400 to 700 feet.

Question 29 asks about the preceding passage as a whole.

29. Upon reviewing the essay and finding that some information has been left out, the writer composes the following sentence incorporating that information:

This technique, also known as "open excavation," became the standard for subway tunneling for nearly sixty years.

If the writer were to add this sentence to the essay, the sentence would most logically be placed at Point:

- A. A.
B. B.
C. C.
D. D.

PASSAGE III

Diego Rivera: The People's Painter

In the 1920s, Mexican artist Diego Rivera (1886–1957) practiced³⁰ the art of painting frescoes, large murals done on fresh plaster. Rivera's frescoes appeared on the outside walls of buildings in Mexico City, in plain sight of any passerby. This brought art out of the elite galleries by catering³¹ to the upper class and literally to the public.

Rivera attracted for his belief controversy that the working class should wield more political power.³² His

dominant artistic subject in his art was as expansive³³ www.crackab.com

than his frescoes:³⁴ the role played by laborers in the past,

present, and future of humanity. One³⁵ of his frescoes depict a progression through time and can be read as time lines from left to right. For example, on the left side of a fresco, there might be field workers hunched over in fatigue and surrounded by the tools of their trade. On the right side, after they have moved through history.³⁶ The same workers stand tall, radiating strength and confidence. Such

empowerment of the worker were to be the bright future³⁷ Rivera envisioned for all the workers of the world.

30. The writer wants to suggest that the art of the fresco had been in decline previous to Rivera. Which choice best accomplishes that goal?
- F. NO CHANGE
G. engaged in
H. influenced
J. revived
31. A. NO CHANGE
B. that catered
C. while catering
D. and catered
32. F. NO CHANGE
G. Rivera should wield more political power for his belief that controversy attracted the working class.
H. Rivera for his controversy attracted belief that the working class should wield more political power.
J. Rivera attracted controversy for his belief that the working class should wield more political power.
33. A. NO CHANGE
B. that he was interested in
C. that he focused on
D. DELETE the underlined portion.
34. F. NO CHANGE
G. then
H. as
J. if
35. A. NO CHANGE
B. Many
C. Each
D. Any one
36. F. NO CHANGE
G. history; the
H. history, the
J. history—the
37. A. NO CHANGE
B. if it were
C. was
D. if it was



Rivera received various prestigious commissions while he was in the United States. In the 1930s, he was commissioned by the Ford Motor Company to paint a twenty-seven-panel fresco in the Detroit Institute of Arts. The fresco, *Detroit Industry*, portrays some of the varied groups that shaped American culture and constituted its workforce. The central panel on the north wall shows the manufacture of a 1932 Ford V-8 engine, when the central panel on the south wall shows the production of this same car's exterior. Smaller panels depicting workers in a variety of other Detroit industries. 42 The fresco is a dynamic work because, by capturing the energy, humanity, and collective achievement of the Detroit workers, celebrates all working men and women. However, Rivera considered it the greatest achievement of his career.

38. F. NO CHANGE
G. various, prestigious,
H. various, and prestigious
J. various and prestigious,
39. If the underlined phrase were deleted, the sentence would primarily lose a detail that:
A. repeats information found elsewhere in the sentence.
B. is necessary for the sentence to be grammatically complete.
C. provides new and relevant information to the sentence.
D. is ambiguous and unnecessary to the sentence.
40. F. NO CHANGE
G. since
H. thus
J. and
41. A. NO CHANGE
B. depict
C. depicting some
D. had depicted
42. The writer is thinking of adding the following phrase to the end of the preceding sentence (changing the period after *industries* to a comma):
such as medicine, pharmaceuticals, and chemicals.
Should the writer make this addition there?
F. Yes, because it offers relevant examples that help to specify a broad term.
G. Yes, because it helps explain how the panels were physically constructed.
H. No, because it provides a sampling of industries rather than a full listing.
J. No, because it digresses from the main point of the sentence.
43. A. NO CHANGE
B. that,
C. while,
D. that was,
44. F. NO CHANGE
G. Despite this,
H. Regardless,
J. DELETE the underlined portion.

PASSAGE IV

After All These Years

[1]

[1] I met Joan, the person who would be my best friend for the next twenty years, the first morning I played outside my family's new California home. [2] I was five years old. [3] We became inseparable childhood friends, and we remained close, even though we attended different high schools and colleges.

[2]

Joan enjoyed jogging and painting cityscapes. I loved hiking trips and writing. We shared an appreciation of the outdoors and a passion for our creative work. More importantly though we enjoyed being together. Through our history of shared experiences, we formed a rare understanding of each other.

[3]

[1] Last February, I had to travel to Fairbanks, Alaska, for my work. [2] Though we had rarely spoken to each other in fifteen years, when I called Joan to suggest a meeting, her voice sounded wonderfully familiar.

45. A. NO CHANGE
B. close, yet even
C. close; even
D. close. Even

46. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. cityscapes, while I
G. cityscapes; I
H. cityscapes. I, on the other hand,
J. cityscapes I

47. A. NO CHANGE
B. important though
C. importantly, though,
D. important, though

48. Which choice would best express the narrator's positive reaction to speaking with Joan and the narrator's fondness for her friend?
F. NO CHANGE
G. she said that she would rearrange her schedule so that we could meet.
H. she told me that she immediately recognized my voice.
J. her quick words and the sound of her laugh surprised me.

[3] Through my parents, whom were still in touch with

49

Joan's father, I learned that Joan was currently living in

50

Fairbanks. 51

[4]

I parked my rental car in downtown Fairbanks,
and to keep the battery from freezing, I plugged the
engine into an electrical outlet in the parking lot
so the battery would stay warm. It was twenty below

52

zero that afternoon, and the sky shone with a pale gray

53

light. 54 I called Joan from a pay phone. She soon met
me on a street corner that was close to her art studio.

[5]

As we walked upstairs to her studio,
we slipped into our familiar habits, talking about
the people in our lives and our work. We talked just
as easily as we had in the past, when we would sit
in the field behind Joan's house atop the rabbit hutch
and discuss our friends and our hopes for the future.

55

56

49. A. NO CHANGE
B. who
C. whose
D. which

50. F. NO CHANGE
G. we
H. they
J. he

51. Which of the following sequences of sentences makes Paragraph 3 most logical?
A. NO CHANGE
B. 1, 3, 2
C. 2, 1, 3
D. 3, 2, 1

52. F. NO CHANGE
G. located in the downtown area of the city.
H. so the battery would continue to work properly despite the cold weather.
J. DELETE the underlined portion and end the sentence with a period.

53. Which of the following alternatives to the underlined portion would NOT be acceptable?
A. was glowing
B. glowed
C. shined
D. shoned

54. If the writer were to delete the preceding sentence, the essay would primarily lose:
F. an indication of the narrator's response to the weather conditions in Fairbanks.
G. a detailed analysis of why the narrator had to plug the car engine into an electrical outlet.
H. descriptive details that help set the scene of the narrator's meeting with Joan.
J. unnecessary details that repeat information given earlier in the paragraph.

55. A. NO CHANGE
B. fell upon
C. dropped by
D. returned with

56. F. NO CHANGE
G. in the field atop the rabbit hutch behind Joan's house
H. atop the rabbit hutch in the field behind Joan's house
J. behind Joan's house in the field atop the rabbit hutch



[6]

When I saw Joan's new paintings, I immediately remembered her distinct way of emphasizing shadows and light. I remembered everything about her: how she would get so absorbed in her work that she'd forget to eat, how
57

she disliked talking in the morning, how she was firm in
58

her decisions. The years of separation had not affected the
59
heart of our connection, our friendship.
59

57. Which of the following alternatives to the underlined portion would NOT be acceptable?
- A. engrossed in
 - B. acquired by
 - C. immersed in
 - D. engaged in
58. Which of the following alternatives to the underlined portion would NOT be acceptable?
- F. with
 - G. regarding
 - H. along
 - J. about
59. Given that all the choices are true, which one would best conclude this essay by effectively summarizing its main idea?
- A. NO CHANGE
 - B. Sadly, I realized that although we might be able to meet once a year, Joan and I would probably never again live in the same city.
 - C. Even though we had followed different interests, I was glad to know that both Joan and I had been able to devote time to our creative work.
 - D. As a result of the time we spent together when we were very young, I'll always remember Joan.

Question 60 asks about the preceding passage as a whole.

60. Upon reviewing the essay and finding that some information has been left out, the writer composes the following sentence incorporating that information:
- Yet, despite such strong ties, we moved far apart as adults and lost touch.
- This sentence would most logically be placed:
- F. after Sentence 2 in Paragraph 1.
 - G. at the end of Paragraph 2.
 - H. at the end of Paragraph 4.
 - J. after the first sentence in Paragraph 6.

PASSAGE V

Three Stars, Many Stories

Many thousands of years ago, people around the world began attaching different stories to the stars in the
61

night sky. The Sun sets gradually the images of a winged
62
horse, a drinking gourd, a heartbroken hero appear in lights overhead. In some cases, a pattern of stars may represent a simple object that has meaning in day-to-day life. In other cases, the pattern, or constellation, may be
63

a figure with a different kind of meaning.
64

Three bright stars that I've read about have acquired
65

significance for many viewers around the globe. In some
66

agricultural parts of Japan, for instance, these three stars
67
are commonly referred to as *Karasuki* and represent a

three-pronged plow. It's awesome that in other parts of
68
Japan, the same three stars appear in a constellation

61. A. NO CHANGE
B. stories, which they connected to
C. stories, to which they related to
D. stories because of
62. F. NO CHANGE
G. sets, gradually,
H. sets, and gradually
J. setting gradually
63. A. NO CHANGE
B. pattern, or constellation
C. pattern or constellation,
D. pattern or constellation:
64. Given that all the choices are true, which one ends this paragraph with the clearest allusion to *Orion*, as the constellation is described later in the essay?
F. NO CHANGE
G. that is interesting but hard to see without a telescope.
H. who plays a dramatic role in a myth that has been told and retold for centuries.
J. that is also represented in the night sky once the Sun has set and the stars emerge.
65. Given that all the choices are true, which one offers visual information about the stars as they appear in modern times?
A. NO CHANGE
B. have different names in different cultures
C. formed long before any of us were born
D. together roughly form a straight line
66. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. observers
G. overseers
H. night-sky watchers
J. stargazers
67. Given that all the choices are true, which one provides a detail that has the most direct connection to the information that follows in this sentence?
A. NO CHANGE
B. distant
C. populated
D. historic
68. F. NO CHANGE
G. You'll be amazed to learn that in
H. Consider, if you will, the notion that in
J. In



representing the floor-length sleeve of a woman's kimono.

In still other parts of Japan, this shining trio appears in the center of an hourglass-shaped drum, a *tsuzumi*.

On the other side of the world, the same three stars ⁶⁹has traditionally represented three zebras to the Namaqua people of South Africa. In

the mythology, of the Tswana people of South Africa, ⁷¹these same stars represent three pigs.

[1] *Orion* is the name many Westerners use for a constellation that contains these three stars. [2] In Greek mythology, Orion is a mighty hunter. [3] In the night sky, he carries a bow and arrow and is accompanied by his loyal dogs, *Canis Major* and *Canis Minor*. [4] The three stars form the brilliant belt around the hunter's waist. [5] In the sky with Orion are the animals he used to hunt on Earth—from a small rabbit to a huge bull. [6] The scorpion that, according to myth, killed Orion inhabits the sky as well, but at such a distance because ⁷²it can never sting the hunter again. [7] Even in an age of big-screen televisions, their ⁷³is still no show on Earth as big as the night sky. [8] Stars up there play different roles around the world, their ⁷⁴dazzling careers span thousands of years. 75

69. A. NO CHANGE
B. In Japan's imagination, this
C. In Japan, this
D. This
70. F. NO CHANGE
G. have
H. could of
J. has been
71. A. NO CHANGE
B. mythology of the Tswana people, of South Africa
C. mythology, of the Tswana people, of South Africa
D. mythology of the Tswana people of South Africa,
72. F. NO CHANGE
G. so when
H. this means
J. that
73. A. NO CHANGE
B. they're
C. there
D. but there
74. F. NO CHANGE
G. world, and their
H. world, with
J. world,
75. The writer wants to divide the preceding paragraph into two to create a concluding paragraph that is free of direct references to a specific culture's view of the three stars. The best place to begin the new paragraph would be at the beginning of Sentence:
A. 4.
B. 5.
C. 6.
D. 7.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. The weekly fee for staying at the Pleasant Lake Campground is \$20 per vehicle and \$10 per person. Last year, weekly fees were paid for v vehicles and p persons. Which of the following expressions gives the total amount, in dollars, collected for weekly fees last year?
 - A. $20v + 10p$
 - B. $20p + 10v$
 - C. $10(v + p)$
 - D. $30(v + p)$
 - E. $10(v + p) + 20p$
2. If $r = 9$, $b = 5$, and $g = -6$, what does $(r + b - g)(b + g)$ equal?
 - F. -20
 - G. -8
 - H. 8
 - J. 19
 - K. 20
3. A copy machine makes 60 copies per minute. A second copy machine makes 80 copies per minute. The second machine starts making copies 2 minutes after the first machine starts. Both machines stop making copies 8 minutes after the first machine started. Together, the 2 machines made how many copies?
 - A. 480
 - B. 600
 - C. 680
 - D. 720
 - E. 960
4. Marlon is bowling in a tournament and has the highest average after 5 games, with scores of 210, 225, 254, 231, and 280. In order to maintain this exact average, what *must* be Marlon's score for his 6th game?
 - F. 200
 - G. 210
 - H. 231
 - J. 240
 - K. 245
5. Joelle earns her regular pay of \$7.50 per hour for up to 40 hours of work in a week. For each hour over 40 hours of work in a week, Joelle is paid $1\frac{1}{2}$ times her regular pay. How much does Joelle earn for a week in which she works 42 hours?
 - A. \$126.00
 - B. \$315.00
 - C. \$322.50
 - D. \$378.00
 - E. \$472.50
6. Which of the following mathematical expressions is equivalent to the verbal expression "A number, x , squared is 39 more than the product of 10 and x " ?
 - F. $2x = 39 + 10x$
 - G. $2x = 39x + 10x$
 - H. $x^2 = 39 - 10x$
 - J. $x^2 = 39 + x^{10}$
 - K. $x^2 = 39 + 10x$
7. If $9(x - 9) = -11$, then $x = ?$
 - A. $-\frac{92}{9}$
 - B. $-\frac{20}{9}$
 - C. $-\frac{11}{9}$
 - D. $-\frac{2}{9}$
 - E. $\frac{70}{9}$



8. Discount tickets to a basketball tournament sell for \$4.00 each. Enrico spent \$60.00 on discount tickets, \$37.50 less than if he had bought the tickets at the regular price. What was the regular ticket price?
- F. \$ 2.50
G. \$ 6.40
H. \$ 6.50
J. \$ 7.50
K. \$11.00
9. The expression $(3x - 4y^2)(3x + 4y^2)$ is equivalent to:
- A. $9x^2 - 16y^4$
B. $9x^2 - 8y^4$
C. $9x^2 + 16y^4$
D. $6x^2 - 16y^4$
E. $6x^2 - 8y^4$
10. A rectangle has an area of 32 square feet and a perimeter of 24 feet. What is the shortest of the side lengths, in feet, of the rectangle?
- F. 1
G. 2
H. 3
J. 4
K. 8
11. In $\triangle ABC$, the sum of the measures of $\angle A$ and $\angle B$ is 47° . What is the measure of $\angle C$?
- A. 47°
B. 86°
C. 94°
D. 133°
E. 143°
12. In the school cafeteria, students choose their lunch from 3 sandwiches, 3 soups, 4 salads, and 2 drinks. How many different lunches are possible for a student who chooses exactly 1 sandwich, 1 soup, 1 salad, and 1 drink?
- F. 2
G. 4
H. 12
J. 36
K. 72
13. For 2 consecutive integers, the result of adding the smaller integer and triple the larger integer is 79. What are the 2 integers?
- A. 18, 19
B. 19, 20
C. 20, 21
D. 26, 27
E. 39, 40
14. A function $f(x)$ is defined as $f(x) = -8x^2$. What is $f(-3)$?
- F. -72
G. 72
H. 192
J. -576
K. 576
15. If $3^x = 54$, then which of the following must be true?
- A. $1 < x < 2$
B. $2 < x < 3$
C. $3 < x < 4$
D. $4 < x < 5$
E. $5 < x$
16. What is the least common multiple of 70, 60, and 50?
- F. 60
G. 180
H. 210
J. 2,100
K. 210,000
17. Hot Shot Electronics is designing a packing box for its new line of Acoustical Odyssey speakers. The box is a rectangular prism of length 45 centimeters, width 30 centimeters, and volume 81,000 cubic centimeters. What is the height, in centimeters, of the box?
- A. 75
B. 60
C. 48
D. 27
E. 18
18. Four points, A , B , C , and D , lie on a circle having a circumference of 15 units. B is 2 units counterclockwise from A . C is 5 units clockwise from A . D is 7 units clockwise from A and 8 units counterclockwise from A . What is the order of the points, starting with A and going clockwise around the circle?
- F. A, B, C, D
G. A, B, D, C
H. A, C, B, D
J. A, C, D, B
K. A, D, C, B
19. A group of cells grows in number as described by the equation $y = 16(2)^t$, where t represents the number of days and y represents the number of cells. According to this formula, how many cells will be in the group at the end of the first 5 days?
- A. 80
B. 160
C. 400
D. 512
E. 1,280



20. The length of a rectangle is 3 times the length of a smaller rectangle. The 2 rectangles have the same width. The area of the smaller rectangle is A square units. The area of the larger rectangle is kA square units. Which of the following is the value of k ?

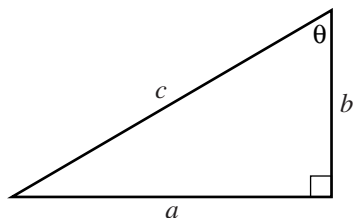
F. $\frac{1}{9}$
 G. $\frac{1}{3}$
 H. 1
 J. 3
 K. 9

21. $(a + 2b + 3c) - (4a + 6b - 5c)$ is equivalent to:

A. $-4a - 8b - 2c$
 B. $-4a - 4b + 8c$
 C. $-3a + 8b - 2c$
 D. $-3a - 4b - 2c$
 E. $-3a - 4b + 8c$

22. The dimensions of the right triangle shown below are given in feet. What is $\sin \theta$?

F. $\frac{a}{b}$
 G. $\frac{a}{c}$
 H. $\frac{b}{c}$
 J. $\frac{b}{a}$
 K. $\frac{c}{a}$



23. In a basketball passing drill, 5 basketball players stand evenly spaced around a circle. The player with the ball (the passer) passes it to another player (the receiver). The receiver cannot be the player to the passer's immediate right or left and cannot be the player who last passed the ball. A designated player begins the drill as the first passer. This player will be the receiver for the first time on which pass of the ball?

A. 4th
 B. 5th
 C. 6th
 D. 10th
 E. 24th

www.actexam.net

24. Lines p and n lie in the standard (x,y) coordinate plane. An equation for line p is $y = 0.12x + 3,000$. The slope of line n is 0.1 greater than the slope of line p . What is the slope of line n ?

F. 0.012
 G. 0.02
 H. 0.22
 J. 1.2
 K. 300

25. The expression $-8x^3(7x^6 - 3x^5)$ is equivalent to:

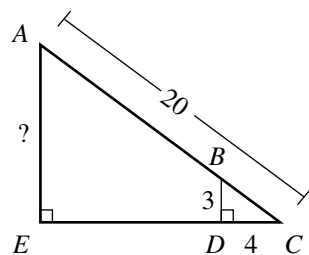
A. $-56x^9 + 24x^8$
 B. $-56x^9 - 24x^8$
 C. $-56x^{18} + 24x^{15}$
 D. $-56x^{18} - 24x^{15}$
 E. $-32x^4$

26. $-3|-6 + 8| = ?$

F. -42
 G. -6
 H. -1
 J. 6
 K. 42

27. In right triangle $\triangle ACE$ below, $\overline{BD}$ is parallel to $\overline{AE}$, and $\overline{BD}$ is perpendicular to $\overline{EC}$ at D . The length of $\overline{AC}$ is 20 feet, the length of $\overline{BD}$ is 3 feet, and the length of $\overline{CD}$ is 4 feet. What is the length, in feet, of $\overline{AE}$?

A. 10
 B. 12
 C. 15
 D. 16
 E. 17



28. As part of a lesson on motion, students observed a cart rolling at a constant rate along a straight line. As shown in the chart below, they recorded the distance, y feet, of the cart from a reference point at 1-second intervals from $t = 0$ seconds to $t = 5$ seconds.

t	0	1	2	3	4	5
y	14	19	24	29	34	39

Which of the following equations represents this data?

F. $y = t + 14$
 G. $y = 5t + 9$
 H. $y = 5t + 14$
 J. $y = 14t + 5$
 K. $y = 19t$



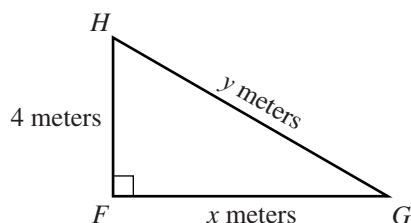
29. The inequality $6(x + 2) > 7(x - 5)$ is equivalent to which of the following inequalities?

A. $x < -23$
 B. $x < 7$
 C. $x < 17$
 D. $x < 37$
 E. $x < 47$

30. The sides of a square are 3 cm long. One vertex of the square is at $(2,0)$ on a square coordinate grid marked in centimeter units. Which of the following points could also be a vertex of the square?

F. $(-4, 0)$
 G. $(0, 1)$
 H. $(1, -1)$
 J. $(4, 1)$
 K. $(5, 0)$

31. For $\triangle FGH$, shown below, which of the following is an expression for y in terms of x ?

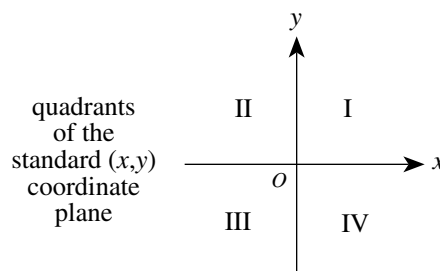


A. $x + 4$
 B. $\sqrt{x^2 + 4}$
 C. $\sqrt{x^2 + 8}$
 D. $\sqrt{x^2 - 16}$
 E. $\sqrt{x^2 + 16}$

32. A bag contains 12 red marbles, 5 yellow marbles, and 15 green marbles. How many additional red marbles must be added to the 32 marbles already in the bag so that the probability of randomly drawing a red marble is $\frac{3}{5}$?

F. 13
 G. 18
 H. 28
 J. 32
 K. 40

33. What are the quadrants of the standard (x,y) coordinate plane below that contain points on the graph of the equation $4x - 2y = 8$?



A. I and III only
 B. I, II, and III only
 C. I, II, and IV only
 D. I, III, and IV only
 E. II, III, and IV only

34. The graph of $y = -5x^2 + 9$ passes through $(1, 2a)$ in the standard (x,y) coordinate plane. What is the value of a ?

F. 2
 G. 4
 H. 7
 J. -1
 K. -8

35. Jerome, Kevin, and Seth shared a submarine sandwich. Jerome ate $\frac{1}{2}$ of the sandwich, Kevin ate $\frac{1}{3}$ of the sandwich, and Seth ate the rest. What is the ratio of Jerome's share to Kevin's share to Seth's share?

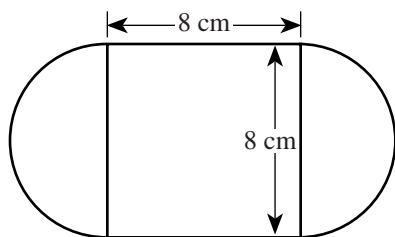
A. 2:3:6
 B. 2:6:3
 C. 3:1:2
 D. 3:2:1
 E. 6:3:2

36. A particular circle in the standard (x,y) coordinate plane has an equation of $(x - 5)^2 + y^2 = 38$. What are the radius of the circle, in coordinate units, and the coordinates of the center of the circle?

	radius	center
F.	$\sqrt{38}$	$(5, 0)$
G.	19	$(5, 0)$
H.	38	$(5, 0)$
J.	$\sqrt{38}$	$(-5, 0)$
K.	19	$(-5, 0)$

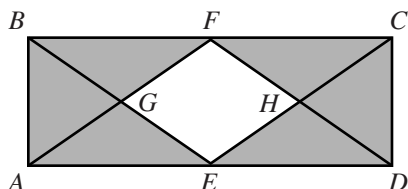


37. The figure below consists of a square and 2 semicircles, with dimensions as shown. What is the outside perimeter, in centimeters, of the figure?



- A. $8 + 8\pi$
 B. $16 + 8\pi$
 C. $16 + 16\pi$
 D. $32 + 8\pi$
 E. $32 + 16\pi$

38. In the figure below, points E and F are the midpoints of sides $\overline{AD}$ and $\overline{BC}$ of rectangle $ABCD$, point G is the intersection of $\overline{AF}$ and $\overline{BE}$, and point H is the intersection of $\overline{CE}$ and $\overline{DF}$. The interior of $ABCD$ except for the interior of $EGFH$ is shaded. What is the ratio of the area of $EGFH$ to the area of the shaded region?



- F. 1:2
 G. 1:3
 H. 1:4
 J. 1:6
 K. Cannot be determined from the given information

39. The coordinates of the endpoints of $\overline{CD}$, in the standard (x,y) coordinate plane, are $(-4,-2)$ and $(14,2)$. What is the x -coordinate of the midpoint of $\overline{CD}$?

- A. 0
 B. 2
 C. 5
 D. 9
 E. 10

40. What is the surface area, in square inches, of an 8-inch cube?

- F. 512
 G. 384
 H. 320
 J. 256
 K. 192

41. The equations below are linear equations of a system where a , b , and c are positive integers.

$$\begin{aligned} ay + bx &= c \\ ay - bx &= c \end{aligned}$$

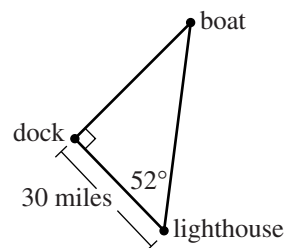
Which of the following describes the graph of at least 1 such system of equations in the standard (x,y) coordinate plane?

- I. 2 parallel lines
 II. 2 intersecting lines
 III. A single line

- A. I only
 B. II only
 C. III only
 D. I or II only
 E. I, II, or III

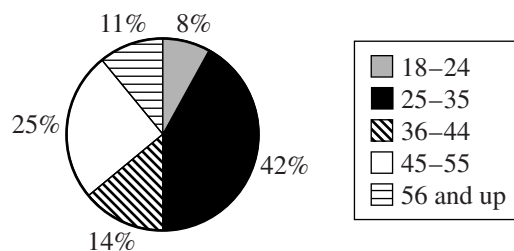
42. According to the measurements given in the figure below, which of the following expressions gives the distance, in miles, from the boat to the dock?

- F. $30 \tan 52^\circ$
 G. $30 \cos 52^\circ$
 H. $30 \sin 52^\circ$
 J. $\frac{30}{\cos 52^\circ}$
 K. $\frac{30}{\sin 52^\circ}$



43. The circle graph below shows the distribution of registered voters, by age, for a community. Registered voters are randomly selected from this distribution to be called for jury duty. What are the odds (in the age range: not in the age range) that the first person called for jury duty is in the age range of 25–35 years?

Distribution of Registered Voters by Age

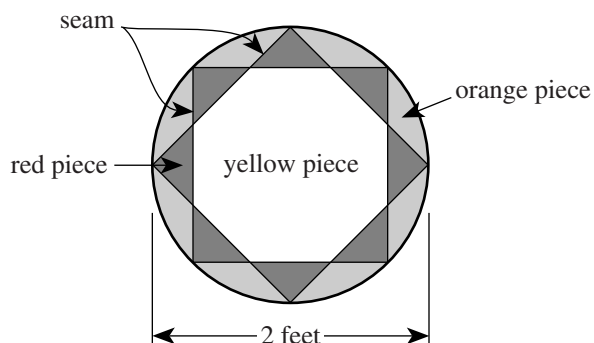


- A. 1:3
 B. 7:8
 C. 7:43
 D. 21:29
 E. 42:25



Use the following information to answer questions 44–46.

The figure below shows the design of a circular stained-glass panel on display at Hopewell's Antique Shop. Seams separate the pieces of the panel. All red triangular pieces shown are congruent and have a common vertex with each adjoining triangular piece. The 2 squares shown are inscribed in the circle. The diameter of the panel is 2 feet.



44. The design of the stained-glass panel has how many lines of symmetry in the plane of the panel?

F. 2
G. 4
H. 8
J. 16
K. Infinitely many

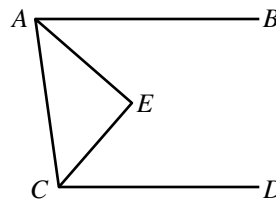
45. What is the area of the stained-glass panel, to the nearest 0.1 square foot?

A. 3.1
B. 4.0
C. 6.2
D. 8.0
E. 12.6

46. Kaya wants to install a new circular stained-glass window in her living room. The design of the window will be identical to that of the panel. The diameter of the new window will be 75% longer than the diameter of the panel. The new window will be how many feet in diameter?

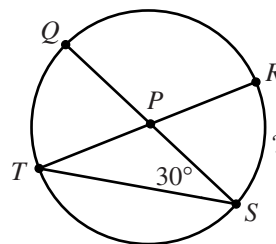
F. 1.50
G. 2.50
H. 2.75
J. 3.50
K. 4.00

47. In the figure below, $\overline{AB} \parallel \overline{CD}$, $\overline{AE}$ bisects $\angle BAC$, and $\overline{CE}$ bisects $\angle ACD$. If the measure of $\angle BAC$ is 82° , what is the measure of $\angle AEC$?



- A. 86°
B. 88°
C. 90°
D. 92°
E. Cannot be determined from the given information

48. In the circle shown below, chords $\overline{TR}$ and $\overline{QS}$ intersect at P , which is the center of the circle, and the measure of $\angle PST$ is 30° . What is the degree measure of minor arc $\widehat{RS}$?



- F. 30°
G. 45°
H. 60°
J. 90°
K. Cannot be determined from the given information

49. For what value of a would the following system of equations have an infinite number of solutions?

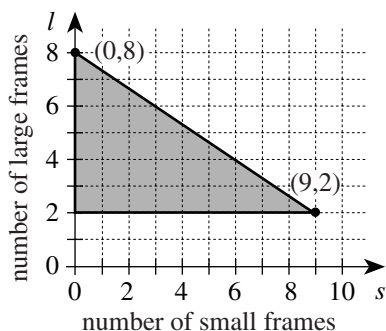
$$\begin{aligned} 2x - y &= 8 \\ 6x - 3y &= 4a \end{aligned}$$

- A. 2
B. 6
C. 8
D. 24
E. 32



Use the following information to answer questions 50–52.

Marcia makes and sells handcrafted picture frames in 2 sizes: small and large. It takes her 2 hours to make a small frame and 3 hours to make a large frame. The shaded triangular region shown below is the graph of a system of inequalities representing weekly constraints Marcia has in making the frames. For making and selling s small frames and l large frames, Marcia makes a profit of $30s + 70l$ dollars. Marcia sells all the frames she makes.



50. The weekly constraint represented by the horizontal line segment containing $(9,2)$ means that each week Marcia makes a minimum of:

F. 2 large frames.
G. 9 large frames.
H. 2 small frames.
J. 9 small frames.
K. 11 small frames.

51. For every hour that Marcia spends making frames in the second week of December each year, she donates \$3 from that week's profit to a local charity. This year, Marcia made 4 large frames and 2 small frames in that week. Which of the following is closest to the percent of that week's profit Marcia donated to the charity?

A. 6%
B. 12%
C. 14%
D. 16%
E. 19%

52. What is the maximum profit Marcia can earn from the picture frames she makes in 1 week?

F. \$410
G. \$460
H. \$540
J. \$560
K. \$690

53. The *determinant* of a matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ equals $ad - cb$.

What must be the value of x for the matrix $\begin{bmatrix} x & 8 \\ x & x \end{bmatrix}$ to

have a determinant of -16 ?

A. -4
B. -2
C. $-\frac{8}{5}$
D. $\frac{8}{3}$
E. 4

54. A formula for finding the value, A dollars, of P dollars invested at $i\%$ interest compounded annually for n years is $A = P(1 + 0.01i)^n$. Which of the following is an expression for P in terms of i , n , and A ?

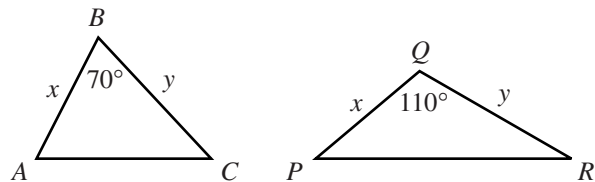
F. $A - 0.01i^n$
G. $A + 0.01i^n$
H. $\left(\frac{A}{1 + 0.01i}\right)^n$
J. $\frac{A}{(1 - 0.01i)^n}$
K. $\frac{A}{(1 + 0.01i)^n}$

55. If x and y are real numbers such that $x > 1$ and $y < -1$, then which of the following inequalities *must* be true?

A. $\frac{x}{y} > 1$
B. $|x|^2 > |y|$
C. $\frac{x}{3} - 5 > \frac{y}{3} - 5$
D. $x^2 + 1 > y^2 + 1$
E. $x^{-2} > y^{-2}$



56. Triangles $\triangle ABC$ and $\triangle PQR$ are shown below. The given side lengths are in centimeters. The area of $\triangle ABC$ is 30 square centimeters. What is the area of $\triangle PQR$, in square centimeters?



- F. 15
G. 19
H. 25
J. 30
K. 33
57. Triangle $\triangle ABC$ is shown in the figure below. The measure of $\angle A$ is 40° , $AB = 18$ cm, and $AC = 12$ cm. Which of the following is the length, in centimeters, of $\overline{BC}$?
- (Note: For a triangle with sides of length a , b , and c opposite angles $\angle A$, $\angle B$, and $\angle C$, respectively, the law of sines states $\frac{\sin \angle A}{a} = \frac{\sin \angle B}{b} = \frac{\sin \angle C}{c}$ and the law of cosines states $c^2 = a^2 + b^2 - 2ab \cos \angle C$.)
-
- A. $12 \sin 40^\circ$
B. $18 \sin 40^\circ$
C. $\sqrt{18^2 - 12^2}$
D. $\sqrt{12^2 + 18^2}$
E. $\sqrt{12^2 + 18^2 - 2(12)(18) \cos 40^\circ}$
58. What is the sum of the first 4 terms of the arithmetic sequence in which the 6th term is 8 and the 10th term is 13 ?
- F. 10.5
G. 14.5
H. 18
J. 21.25
K. 39.5
59. In the equation $x^2 + mx + n = 0$, m and n are integers. The *only* possible value for x is -3 . What is the value of m ?
- A. 3
B. -3
C. 6
D. -6
E. 9
60. The solution set of which of the following equations is the set of real numbers that are 5 units from -3 ?
- F. $|x + 3| = 5$
G. $|x - 3| = 5$
H. $|x + 5| = 3$
J. $|x - 5| = 3$
K. $|x + 5| = -3$

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is accompanied by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the short story “From Aboard the Night Train” by Kimberly M. Blaeser (©1993 by Kimberly M. Blaeser), which appeared in *Earth Song, Sky Spirit: Short Stories of the Contemporary Native American Experience*.

The passage begins with a female narrator traveling to her hometown.

The moon gives some light and I can make out the contours of the land, see the faint reflection in the lakes and ponds we pass. Several times I see or imagine I see glowing eyes staring back at me from a patch of woods
5 beside the track. When we pass through the tiny towns, I try to read their signs, catch their names from their water towers or grain elevators. Occasionally the train stops at . . . Portage . . . Winona . . . Red Wing.

In my sleeping compartment, watching the night
10 countryside, so much world rolls by my window. Like a voyeur I watch the various reunion scenes. I feel these scenes add up to something, some meaning or lesson about all life, and I try to put it into words for myself but find I can’t. I finally give up, roll over, go to sleep,
15 and dream.

But now I am awake, keeping my vigil over the Midwest’s pastoral kingdom. Chicago, even Minneapolis seems a long way away. A few hours later, still in the deep night hours, the train arrives at my stop,
20 Detroit Lakes, Minnesota, the closest I can get to my destination.

Suddenly, as I descend the two steps from the train, the porter hands me into one of the reunion scenes. “Hi, honey, how was the trip? Did you get any
25 sleep?” “A little. Been waiting long?” “Long enough to beat your dad in two games of cribbage . . .” Glancing back at the train windows, I imagine I am looking into eyes hidden behind mirrored sunglasses.

* * *

I think about progress a lot in the next few days and about what passes for progress. Nightly we walk about town, talk marriages and funerals, then sit on the newly installed benches on Main Street. Together we
30 www.actexam.net memories the town as it was twenty or twenty-five years ago. We remember the little Model

Meat Market and the old Pioneer office. We rebuild the Landmark Hotel, take down the vinyl fronts from the grocery store, change the light posts, the awnings, the names of the current businesses. I put back the old
40 depot, you the corner funeral home. But soon we are distracted and leave things half constructed when we begin to add the people, what’s-his-name, the square dance caller; Ed, the fire chief; and Lydia, the town’s best gossip. On the walk back home, we have begun to
45 list very specific things, which is the closest we get to the intangibles: the rental meat lockers, the four-digit telephone numbers, the free ice cream during dairy month.

Late at night in my old bed, I listen to the night
50 sounds of the house and fall asleep counting the changes that have come to my little hometown: The park is off limits after dark now, the football field is fenced in, one-hour photo has come to town along with a tanning salon and a pizza parlor. The dry goods store
55 is gone, the dairy, long gone. People lock their houses now more than once a year when the carnival comes to town. But all of these changes pale in comparison to what has replaced the bait shop, the used car lot, and Mr. Morton’s small farm, what has sprung up on Highway
60 way 59 at the edge of town: Las Vegas-style gambling.

* * *

Taking the train back, I decide to put on pajamas and crawl under the sheets, hoping to trick myself into a good night’s sleep. It seems to work. I have slept
65 soundly for several hours, but then the dreams start. I fall in and out of them. But they are not the usual nightmares. I am in a place where folks know you ten, fifteen, twenty years after you’ve left and still see in your face that of your grandfather or aunt or cousin. I know I
70 am home and I feel safe.

I have an early breakfast with a would-be journalist and some ski vacationers who want to talk about election prospects. I merely feign attention. I nod or laugh on cue, while I try to read upside-down a story in
75 the would-be journalist’s newspaper that has caught my eye. It is about the Russian space station and the cosmonaut who had been up in orbit during the takeover attempt and ultimate dissolution of the Soviet Union. After sixteen long months, they are bringing the capsule back. While the train carries me back to my current
80 home and away from my former, I keep thinking about

that poor cosmonaut coming back to find his whole world changed, to find himself a man without a country—at least without the country he left behind.

85 I watch the ten o'clock national news broadcast. I see him emerge from the capsule. I see him try to stand and have his knees buckle. I know they said it was because he hadn't been able to exercise for such a long time, but I wonder if his weak-kneed feeling might not
90 have more to do with what he saw out the window of the space station and with how the world was happening around without him.

1. The point of view from which the passage is told is best described as that of:
 - A. a young adult riding a train through the small towns of the Upper Midwest.
 - B. a young adult preparing to move away from her hometown.
 - C. an adult missing the new home she has established.
 - D. an adult reflecting on the past and pondering the present.
2. The passage contains recurring references to all of the following EXCEPT:
 - F. dreams.
 - G. reunion scenes.
 - H. photographs.
 - J. train trips.
3. The first three paragraphs (lines 1–21) establish all of the following about the narrator EXCEPT that she is:
 - A. passing through a number of towns.
 - B. originally from Chicago.
 - C. traveling by train.
 - D. observant of the landscape.
4. It can reasonably be inferred from the passage that the narrator thinks her hometown has:
 - F. improved significantly over the years.
 - G. made little genuine progress.
 - H. remained about the same as it was years ago.
 - J. a chance of being rebuilt as it used to be.
5. Based on the narrator's account, all of the following were part of the past, rather than the present, in her hometown EXCEPT:
 - A. four-digit phone numbers.
 - B. the fenced-in football field.
 - C. free ice cream during dairy month.
 - D. the depot.
6. According to the narrator, which of the following businesses is relatively new to her hometown?
 - F. The tanning salon
 - G. The bait shop
 - H. The dry goods store
 - J. The used-car lot
7. When the narrator refers to the cosmonaut as "a man without a country" (lines 83–84), she is most likely directly referring to the:
 - A. cosmonaut's feeling that he is now a citizen of space, not the former Soviet Union.
 - B. cosmonaut's unrealized expectation that he will be treated like a hero.
 - C. political transformation that occurred while the cosmonaut was in space.
 - D. sixteen months that the cosmonaut spent in orbit around Earth.
8. Details in the passage most strongly suggest that the people meeting the narrator at the train station include:
 - F. her father.
 - G. her sister.
 - H. a neighbor.
 - J. a journalist.
9. The narrator indicates that the most significant change to her hometown has been the addition of:
 - A. square dancing.
 - B. vinyl storefronts.
 - C. benches on Main Street.
 - D. Las Vegas-style gambling.
10. According to the passage, news reports attributed the cosmonaut's knees buckling to:
 - F. his gratitude at being back on Earth.
 - G. political changes in the world.
 - H. a lack of exercise.
 - J. his dismay at what he had seen from the space station.

Passage II

SOCIAL SCIENCE: This passage is adapted from the article “Green Music in the Rain Forest” by Suzanne Charlé, which appeared in the Fall 2002 *Ford Foundation Report*.

OELA is an acronym based on Portuguese words rather than the English words used in this article. A *luthier* is a maker of stringed musical instruments.

The Amazonian Workshop School for Fabrication of Stringed Instruments (OELA) is a small part of a larger effort to create a sustainable harvest of the great Amazon forest and to give employment to the region’s
5 burgeoning population.

“Few people know that the Amazon is one of the most rapidly urbanizing regions of the world,” observes José Gabriel López, a Ford Foundation program officer in Brazil. The city of Manaus, for example, has grown
10 in the past decade from 850,000 to 1.5 million. “This rural-urban migration and the resultant urban shantytowns stand as living symbols of failed or nonexistent rural development policies,” López says. “In many places, small-scale rural producers have been abandoned—devoid of health and education services, credit,
15 technical assistance and opportunity. What Rubens Gomes, founder of the workshop school, and his colleagues have created in Manaus is hope.”

Gomes knows how to build hope. The school, he
20 notes proudly, is the first to make stringed instruments in the Amazon. And it is the first in all of the Americas to construct instruments exclusively of lumber harvested in an environmentally and socially sustainable manner certified by the Forest Stewardship Council.

“Officially, there are 30 million cubic meters of wood cut in the Amazon annually,” Gomes says. “Twenty million of this is wasted—sawdust, scraps, unwanted wood left to rot. And those are the official
25 numbers. The motive of this school is to transform what is lost into things of value. Many people could do this—but there are no schools teaching carpentry in the Amazon.”

OELA is meant to help fill the void. To graduate, each student must make a stringed instrument. All the
35 guitars are made from certified wood. Gomes explains that traditionally, Brazilian rosewood and ebony were used in the construction of guitars. But because of intense harvesting, these trees are close to extinction. “I’ve been working for years, trying to find Amazon
40 woods that are unknown on the market, that are in plentiful supply and that can be used in instrument making,” Gomes says. He experimented with dozens before he found types that have the right strength and sound. (Like other master luthiers, he can tell by touch-
45 ing the wood whether it will reverberate well.) Once he identified the woods as possible substitutes, he sent them to a laboratory to be tested for the right grain and density. Today, *Brosimum rubescens* is substituted for rosewood, *Aniba canellila* for ebony, and *Protium*
50 species for Brazilian mahogany and cedar. These and

some 25 other undervalued tropical hardwoods have found their way into the luthiers’ workshop, taking the pressure off the better-known woods.

For the past year, master luthier Raúl Lage from
55 the Fernando Ortiz Instrument-Making School of the Cuban Music Institute has been working with the students. There are hurdles, he cautions, a number of them technical. The high humidity in Manaus means that the wood will crack in drier climates unless properly
60 treated. Glue frequently doesn’t hold. These problems are slowly being resolved.

There is also a major obstacle outside the workshop: The resistance of buyers to new woods. Thus far, most of the instruments have been sold to environmentalists, some of whom “adopt” a student by paying his
65 or her tuition; the student’s “project guitar” is then given to the donor as a gift.

There is also the possibility of contract work from outside the Amazon. Gomes’s hopes were raised
70 recently when the president of a well-known guitar company based in Nashville, Tennessee, ordered 15 guitars to be auctioned off for the Rainforest Alliance.

Lage cautions that it will be a long time before any
75 of the students can command a master luthier’s fee. “There is a saying,” Lage says. “Anyone can make one good guitar; it takes a master to make one every time.”

José Lucio do Nascimento Rabelo, director of the technical school, says, “By learning this skill, students
80 come to look at the forest in a new way; there are ways other than logging for plywood and firewood to earn a living, to better the life of the people.” One of the woods being used as a replacement for the precious rosewood, he notes, is typically used to make charcoal.

Such an appreciation for the forest, says Rabelo,
85 could have a huge effect on the survival of the rain forest; some 80 percent of the students come from other parts of the state of Amazonas, and virtually all of them return to their home towns. “Some,” he adds, “go on to
90 become politicians who will have a direct influence on the future of the forest.”

11. Which of the following assumptions would be most critical for a reader to accept in order to agree fully with the author’s claims in the passage?

- A. Shantytowns in the Amazon need to be relocated if the forest is to be saved.
- B. Learning to make consistently good guitars requires access to the best materials available.
- C. Small-scale rural producers in the Amazon can help preserve the forest by being innovative.
- D. Consumers outside of the Amazon can do little to help prevent deforestation.

12. In the context of the passage, the statement “All the guitars are made from certified wood” (lines 34–35) most nearly suggests that Gomes’s workshop:
- F. uses environmentally sustainable woods in its guitars.
 - G. isn’t doing enough to stop unnecessary deforestation in the Amazon.
 - H. has little chance of pleasing both musicians and environmentalists.
 - J. uses only traditional woods in making its guitars.
13. It can most reasonably be inferred from the passage that regarding OELA, the author feels:
- A. skeptical of the workshop’s aims.
 - B. dismayed by the workshop’s low productivity.
 - C. supportive of the workshop’s goals.
 - D. confident that the workshop could be duplicated in other places.
14. The main purpose of the second paragraph (lines 6–18) is to:
- F. draw attention to the Amazon’s tremendous population growth.
 - G. explain the necessity for ventures such as Gomes’s.
 - H. explain the presence of the Ford Foundation in the Amazon.
 - J. justify raising taxes to increase social services in the Amazon.
15. The main function of the fifth paragraph (lines 33–53) is to:
- A. demonstrate the woodworking skills required to be a master luthier.
 - B. explore the limitations of science as compared to intuition.
 - C. outline the scientific reasons why one type of wood cannot be replaced by another.
 - D. show that experiments led to the discovery of good substitutes for rare woods.
16. The passage notes all of the following as problems that the fledgling Amazon guitar industry has experienced EXCEPT that:
- F. glue on the guitars sometimes doesn’t hold.
 - G. the wood used may crack in drier climates.
 - H. woods usable for guitars have become extinct.
 - J. buyers resist guitars made with nontraditional woods.
17. The passage indicates that, as a group, the OELA students may impact the survival of the rain forests because most of them:
- A. care deeply enough about music to spend their lives making musical instruments.
 - B. will return to their homes and spread their environmental knowledge.
 - C. are willing to endure personal hardships in order to use their new skills.
 - D. will have political careers after they return home.
18. In the passage, Gomes indicates that of the wood cut in the Amazon rain forest each year, approximately how much wood is wasted?
- F. One-fourth
 - G. One-third
 - H. One-half
 - J. Two-thirds
19. The passage states that all of the following are woods traditionally used for making stringed instruments EXCEPT:
- A. Aniba canellila.
 - B. rosewood.
 - C. Brazilian mahogany.
 - D. ebony.
20. According to the passage, when an OELA student is “adopted,” he or she receives:
- F. tuition.
 - G. room and board.
 - H. food and clothing.
 - J. a musical instrument.

Passage III

HUMANITIES: This passage is adapted from the article “Finding Philosophy” by Colin McGinn (©2003 by Prospect).

Descartes (line 63) refers to René Descartes (1596–1650), a French mathematician, philosopher, and scientist.

I have been an academic philosopher for the past 30 years. I came from an academically disinclined background in the northeast of England, my relatives being mainly coalminers and other manual workers. I was the first in my family to attend university, and indeed had no thought of it until age 17, when a teacher mentioned it at school. My father had become a successful builder, so we were not materially deprived, and it was expected that I would become some sort of technical worker. The idea that I might one day become a professional philosopher was inconceivable in those days, to me and everyone else. I was simply not living in a place where that kind of thing ever happened; it was far likelier—though still not at all likely—that I would become a pop star (I played drums in a rock band).

The paperback British edition of my memoir *The Making of a Philosopher* has a photograph on the cover of a man sitting on a bench, placed in a grey and listless landscape. He is overlooking the sea on a misty grim day, and the atmosphere is bleak and melancholy. The man, hunched up, immobile, coiled almost, has a pensive posture, as if frozen in thought. This picture is based on a story I tell in the book about sitting on a bench in Blackpool, aged 18, pondering the metaphysical question of how objects relate to their properties. Is an object just the sum total of its properties, a mere coalescence of general features, or does it somehow lie behind its properties, supporting them, a solid peg on which they happen to hang? When I look at an object do I really see the object itself, or just the appearance its properties offer to me? I remember the feeling of fixation that came over me when I thought about these questions—a kind of floating fascination, a still perplexity.

When I look back on this period in my late teens, I recall the harnessing of undirected mental energy by intellectual pursuits. Up until then, my mental energy had gone into things like reading *Melody Maker*, which contained fairly serious articles about pop musicians; I always knew the top 20 off by heart, and studied the articles about drummers intensely, hoping to improve my own technique. I suspect that this kind of swashing mental energy is fairly typical of boys that age. School doesn't seem to connect with it, and it goes off in search of some object of interest, often trivial, sometimes destructive. In my case, it was philosophy that seized that energy and converted it into a passion—though one that took several years to form fully. It is a delicate and fastidious energy that I am speaking of, despite its power, and it will only be satisfied by certain employments, which of course vary from person to person. I had had a similar passion for chemistry when

I was ten, and for butterflies and lizards before that. How to harness such passions to formal education remains a great and unresolved problem.

It was—of course—a teacher who tapped into my formless and fizzing mental energy. Mr Marsh, teacher of divinity, brimmingly Christian, a man with very active eyebrows and sharp enunciation, in love with scholarship (oh, how he relished that word)—it was he who first brought out my inner philosopher. From him I heard of Descartes, locked up in his room, wondering whether anything could really be known beyond his own existence. But what I mainly got from the enthusiastic Mr Marsh was the desire to study. His own passion for study shone through, and he managed to make it seem, if not glamorous, then at least exhilarating—when done the right way and in the right spirit. Pencils and stationery were made to seem like shiny tools, and the pleasure of making one's mark on a blank sheet of paper hymned. Choosing a good spot to study was emphasised. Above all, I learned a very valuable lesson, one that had hitherto escaped me: make notes. Thinking and writing should be indissoluble activities, the hand ministering to the thought, the thought shaped by the hand. Today, if I find myself without pen and paper and thoughts start to arrive, my fingers begin to twitch and I long for those implements of cogitation. With such rudimentary tools you can perform the miracle of turning an invisible thought into a concrete mark, bringing the ethereal interior into the public external world, refining it into something precious and permanent. The physical pleasure of writing, which I find survives in the use of a computer, is something worth dwelling on in matters of education.

21. The passage is best described as being told from the point of view of a philosopher who is:
 - A. discussing metaphysical questions that have troubled philosophers since the time of Descartes.
 - B. presenting in chronological order the key events in his thirty-year professional career.
 - C. reflecting on his own early, developing interest in philosophy and in scholarship generally.
 - D. advising professional educators on how to get more students to study philosophy.
22. Based on the passage, which of the following was most likely the first to engage the author's passionate interest?
 - F. Drumming
 - G. Philosophy
 - H. Chemistry
 - J. Butterflies

23. The main purpose of the last paragraph is to:
- A. reveal the enduring impact of Mr. Marsh's lessons on the author.
 - B. acknowledge that the author came to doubt some of Mr. Marsh's teachings.
 - C. describe a typical class as taught by Mr. Marsh.
 - D. present a biographical sketch of Mr. Marsh.
24. The passage indicates that the man in the book-cover photograph represents:
- F. Descartes, wondering what could be known.
 - G. Mr. Marsh, deep in scholarly thought.
 - H. the author at age seventeen, thinking about enrolling in college.
 - J. the author at age eighteen, contemplating a philosophical issue.
25. The author mentions *Melody Maker*, the top 20, and articles about musicians primarily to suggest that his:
- A. early interest in music has remained with him to the present.
 - B. time spent playing music should instead have been spent reading.
 - C. fascination with pop music and musicians gave focus to his life for a time.
 - D. commitment to study enabled him to perfect his drumming technique.
26. In the third paragraph (lines 36–56), the author most nearly characterizes the energy he refers to as:
- F. potent yet difficult to channel in a constructive way.
 - G. powerful and typically leading to destructive results.
 - H. delicate and inevitably wasted in trivial undertakings.
 - J. gentle yet capable of uniting people who have different interests.
27. Viewed in the context of the passage, the statement in lines 55–56 is most likely intended to suggest that:
- A. schools should require students to take philosophy courses.
 - B. students can become passionate when learning about science in school.
 - C. schools need to keep searching for ways to tap into students' deeply held interests.
 - D. students should resolve to take school courses that interest them.
28. The author calls pen and paper “rudimentary tools” (line 80) as part of his argument that:
- F. the use of computers has made the use of pen and paper obsolete.
 - G. students should become skilled with pen and paper before moving on to better tools.
 - H. while writing with pen and paper can be pleasant, it can also be physically painful.
 - J. although seemingly simple, pen and paper allow people to perform great feats.
29. In the context of the passage, lines 17–23 are best described as presenting images of:
- A. gloom, tension, and fascination.
 - B. anger, bitterness, and betrayal.
 - C. stillness, peacefulness, and relaxation.
 - D. frustration, surprise, and satisfaction.
30. Which of the following does NOT reasonably describe the transition the author presents in lines 80–84?
- F. Precious to commonplace
 - G. Fleeting to permanent
 - H. Invisible to visible
 - J. Private to public

Passage IV

NATURAL SCIENCE: This passage is adapted from *Consider the Eel* by Richard Schweid (©2002 by Richard Schweid).

The known facts, as they are pretty much universally accepted among biologists and naturalists today, are that all the eels in all the rivers of eastern North America and the Caribbean countries, and all the eels in all the rivers of eastern and western Europe, are born in the same area of the Sargasso Sea, a huge area within the Atlantic Ocean, between Bermuda and the Azores, the surface of which is frequently covered with sargassum seaweed. In fact, the word “Sargasso” comes from the Portuguese *sargaço*, meaning seaweed. The sea is about 2,000 miles long and 1,000 miles wide, set off from the surrounding waters of the Atlantic by strong currents. It includes the area known in popular legend as the Bermuda Triangle.

Eels hatch in the Sargasso as larvae and are carried by the ocean currents to either Europe or the United States, a journey that can cover thousands of miles and take years. Where they end up depends on which of two similar species they belong to. Those that are *Anguilla anguilla* invariably wind up in European rivers, and those that enter North American rivers always belong to the species *Anguilla rostrata*. The first person to find eel larvae in the Sargasso Sea was Danish researcher Johannes Schmidt, who published his findings in 1924, after spending 18 years hauling nets in search of eels.

The larvae of both species are shaped like small oval leaves and are called leptocephali. Each leptocephalus begins to assume the form of a tiny eel, called an elver or glass eel, when it gets close to the coasts of either Europe or the Americas. By the time it reaches brackish water, where fresh and salt water mix, it is thin and transparent, hardly bigger than a hair, with a pair of eyes like black dots at one end.

From the estuaries and mouths of rivers, the tiny eels frequently continue upstream, particularly the females, who sometimes go great distances inland. American eels have been found as far up the Mississippi River system as the rivers of Iowa. They keep going upriver until something tells them they’ve reached home, and then they stop. Whatever it is that signals to eels that they are home is definitive—they settle in and live there for as long as 20 years, growing up to a yard long before beginning their journey back to the Sargasso Sea. Scientists determine an eel’s age using a microscope to read the growth rings of its otolith—a small, hard calcium deposit at the base of its skull.

In preparation for the return journey to the Sargasso, sexually mature female eels feed voraciously and change color from the muddy-yellow/green of adult eels, often called yellow eels, to a darker green on top and snow-white on their bellies. At this stage, they are called silver eels. They swim downriver in the fall, on the first leg of their journey to the Sargasso, and when

they reach estuarine waters, they rest, completing their final transformation as silver eels. They will have eaten heavily and will be about 28 percent body fat. They will never eat again, and their digestive systems will atrophy. Their pupils will expand and turn blue. They will need a new kind of sight adapted to the depths of the sea, where there is little light. They will also have to go through a drastic adjustment, via osmosis, in their blood chemistry, to prepare for the tremendous change in water pressure, going from some 14 pounds of freshwater pressure per inch of their bodies to over a ton of ocean pressure per inch. Once they are back in the Sargasso Sea, the females produce eggs for the males to fertilize, and then the adults die.

At least that is what today’s marine biologists and naturalists tell us, although adult eels have never been seen swimming, reproducing, or dying in the Sargasso. In fact, live adult eels have never been seen there at all. The only two adult eels ever reported in the Sargasso Sea were dead, found in the stomachs of other fish. The eel’s migration back to its birthplace and what it actually does when it gets there are assumed to take place far below the water’s surface and, as of the year 2001, were still completely unobserved. However, the eel larvae—the leptocephali that Schmidt found in the Sargasso—were so small that it was certain they had been born recently, and nearby. Such small larvae have never been seen elsewhere, and while eels have never been observed reproducing in the Sargasso, they have never been seen doing so anywhere else either. Scientists believe the larvae hatch out of eggs at a depth of 100–300 yards and rise slowly toward the light at the sea’s surface.

31. One of the main ideas established by the passage is that:

- A. researchers have nearly exhausted their resources after spending decades investigating the Sargasso Sea.
- B. significant gaps still remain in researchers’ understanding of the life cycle of eels.
- C. eels live their entire lives in the Sargasso Sea, but no one has ever seen them there.
- D. female eels turn into silver eels toward the end of their lives.

32. Learning about which of the following had the largest impact on scientists’ current understanding of where eels breed?

- F. The direction in which ocean currents carry eel larvae
- G. The relationship of the yellow eel stage to the silver eel stage
- H. Schmidt’s discovery of eel larvae in the Sargasso Sea
- J. The adult eels found in the stomachs of other fish

33. The main purpose of the fourth paragraph (lines 34–47) is to describe the:
- A. eels' transition from freshwater to the ocean.
 - B. method of determining the age of eels.
 - C. complexity of the Mississippi River system.
 - D. river stage of the eel life cycle.
34. The passage states that the Sargasso Sea is set off from the rest of the Atlantic Ocean by:
- F. the Azores.
 - G. several Caribbean countries.
 - H. powerful winds.
 - J. strong currents.
35. The passage notes that the Sargasso Sea includes:
- A. the eastern North American shore.
 - B. the Bermuda Triangle.
 - C. certain coastal estuaries.
 - D. the mouth of the Mississippi River.
36. As it is used in line 13, the word *popular* most nearly means:
- F. well liked.
 - G. commonly known.
 - H. scientifically accepted.
 - J. most admired.
37. As it is used in line 45, the word *read* most nearly means to:
- A. learn from print.
 - B. observe.
 - C. think about.
 - D. predict.
38. The passage indicates that female eels' pupils expand and turn blue because the eels:
- F. must adapt to see in an environment with much less light than they are used to.
 - G. are about to undergo a change in their blood chemistry.
 - H. no longer need to be able to recognize food sources since they have stopped eating.
 - J. need to be able to recognize the male eels that will fertilize their eggs.
39. The passage most strongly emphasizes that the process of osmosis is necessary for the eels' transition from:
- A. shallower to deeper water.
 - B. feeding to nonfeeding.
 - C. immature to mature form.
 - D. elver to yellow eel.
40. According to the passage, which of the following characteristics of the eel larvae found by Schmidt provided the best evidence that the larvae were hatched in the Sargasso Sea?
- F. Size
 - G. Shape
 - H. Color
 - J. Species

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.



SCIENCE TEST

35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

Finch beak depth (see Figure 1) is an *inheritable* trait (it can be passed from parents to offspring).



Figure 1

Researchers studied the beak depth of 2 species of finches, *Geospiza fortis* and *Geospiza fuliginosa*. Both species live on Island A. *G. fortis* alone lives on Island B, and *G. fuliginosa* alone lives on Island C. For both species, the primary food is seeds. Birds with shallower beaks can efficiently crush and eat only small seeds. Birds with deeper beaks can crush and eat both large and small seeds, but they prefer small seeds.

Study 1

Researchers captured 100 *G. fortis* finches and 100 *G. fuliginosa* finches on Island A. They tagged each bird, measured its beak depth, and released it. Then they calculated the percent of birds having each of the beak depths that had been measured. The researchers followed the same procedures with 100 *G. fortis* finches from Island B and 100 *G. fuliginosa* finches from Island C. The results of this study are shown in Figure 2.

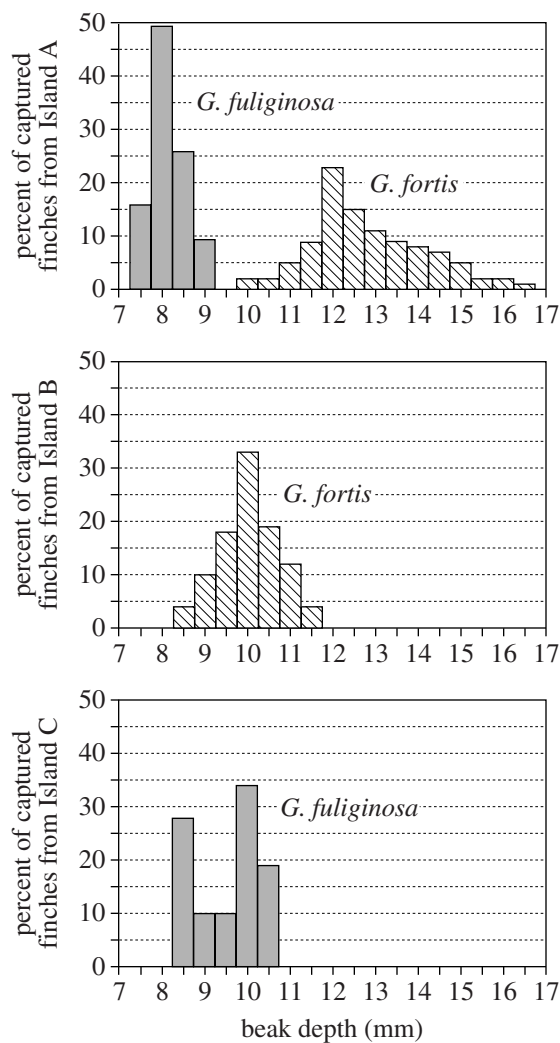


Figure 2

Study 2

After completing Study 1, the researchers returned to Island B each of the next 10 years, from 1976 to 1985. During each visit, the researchers captured at least 50 *G. fortis* finches and measured their beak depths. Then



they calculated the average *G. fortis* beak depth for each of the 10 years. The researchers noted that, during the 10-year period, 3 years were exceptionally dry, and 1 year was very wet (see Figure 3). Small seeds are abundant during wet years. During dry years, all seeds are less abundant, and the average size of the available seeds is larger.

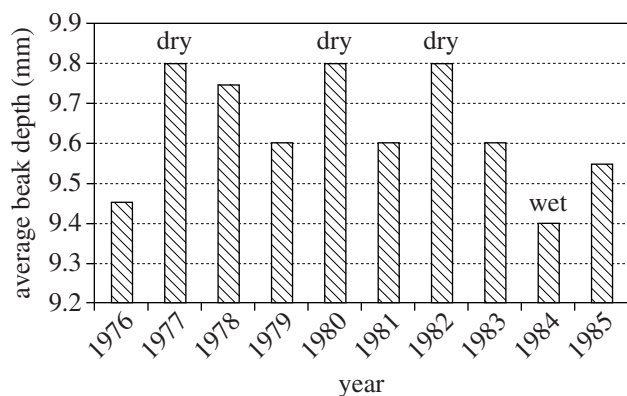


Figure 3

Figures adapted from Neil A. Campbell, Jane B. Reece, and Lawrence G. Mitchell, *Biology*, 5th ed. ©1999 by Benjamin/Cummings.

www.crackab.com

- Based on the results of Study 1, the highest percent of finches on Island B and Island C had a beak depth of:

	Island B	Island C
A.	8 mm	8 mm
B.	9 mm	12 mm
C.	10 mm	8 mm
D.	10 mm	10 mm
- During which of the following years were small seeds likely most abundant on Island B ?
 - 1977
 - 1980
 - 1982
 - 1984
- Study 1 differed from Study 2 in which of the following ways?
 - G. fortis* finches were captured during Study 1 but not during Study 2.
 - G. fuliginosa* finches were captured during Study 1 but not during Study 2.
 - The beak depth of captured birds was measured during Study 1 but not during Study 2.
 - The beak depth of captured birds was measured during Study 2 but not during Study 1.
- It is most likely that the researchers tagged the birds that they captured during Study 1 to:
 - determine how beak depth was affected by rainfall on Island A.
 - determine the average age of each finch population.
 - ensure that the beak depth of each finch was measured multiple times during Study 1.
 - ensure that the beak depth of each finch was measured only once during Study 1.
- Based on the results of Study 2, would a finch with a beak depth of 9.4 mm or a finch with a beak depth of 9.9 mm more likely have had a greater chance of survival during 1977 ?
 - A finch with a beak depth of 9.4 mm, because, on average, the size of available seeds is larger during dry years.
 - A finch with a beak depth of 9.4 mm, because, on average, the size of available seeds is smaller during dry years.
 - A finch with a beak depth of 9.9 mm, because, on average, the size of available seeds is larger during dry years.
 - A finch with a beak depth of 9.9 mm, because, on average, the size of available seeds is smaller during dry years.
- A researcher hypothesized that there would be more variation in the beak depths measured for the *G. fortis* finches when they were forced to compete with another finch species for seeds. Do the results of Study 1 support this hypothesis?
 - Yes; the range of beak depths measured for *G. fortis* finches was greater on Island A than on Island B.
 - Yes; the range of beak depths measured for *G. fortis* finches was greater on Island B than on Island A.
 - No; the range of beak depths measured for *G. fortis* finches was greater on Island A than on Island B.
 - No; the range of beak depths measured for *G. fortis* finches was greater on Island B than on Island A.

Passage II

Substances in the atmosphere, such as Cu^{2+} , Zn^{2+} , Cl^- , and SO_4^{2-} ions, are carried down to Earth's surface by precipitation. This process is known as *wet deposition*. Cu^{2+} and Zn^{2+} ions are put into the atmosphere by high-temperature combustion processes. The presence of Cl^- and SO_4^{2-} ions in the atmosphere can be attributed to road-salt dust and electrical power generation, respectively.

Study 1

A rain gauge, placed on the roof of a 1-story building, at a specific urban site was used to collect precipitation over a 12-month period. At the same time each evening, the amount of precipitation in the rain gauge was recorded, after which the collected precipitation was emptied from the gauge and stored. (Assume no measurable evaporation occurred during any day.) Figure 1 shows the measured monthly precipitation in centimeters.

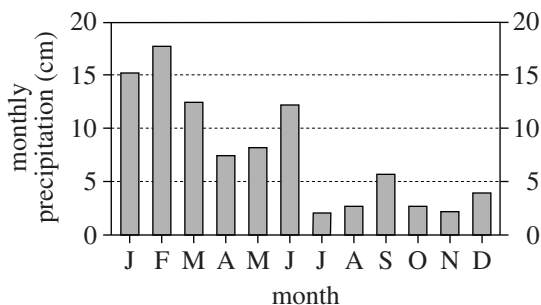


Figure 1

At the end of each month, all the samples collected during that month were mixed, and some of this combined sample was analyzed for the concentrations of Cu^{2+} and Zn^{2+} ions. Using these data, the monthly wet deposition of each substance, in micrograms (μg) per meter², was calculated (see Figure 2).

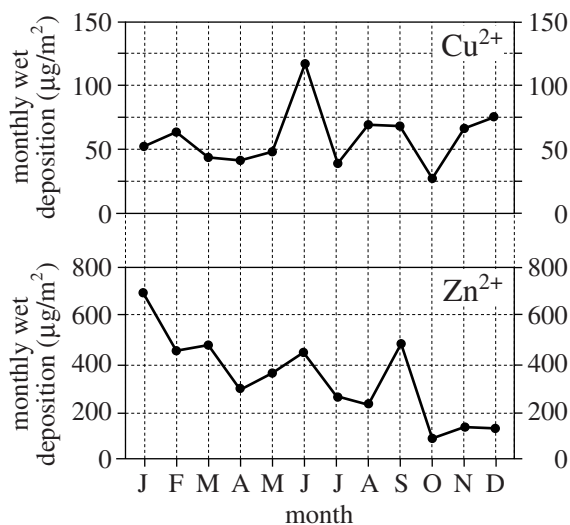


Figure 2

Study 2

Another portion of the combined sample for each month was analyzed for the concentrations of Cl^- and SO_4^{2-} ions. Using these data, the monthly wet deposition of each substance, in milliequivalents (meq) per m², was calculated (see Figure 3).

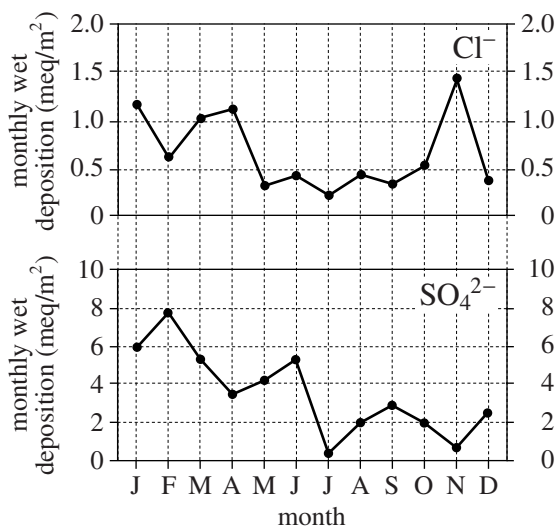


Figure 3

Study 3

The *annual* wet deposition of Cu^{2+} and of Zn^{2+} for the 12-month period, in $\mu\text{g}/\text{m}^2$, was calculated for the urban site (the source of the Cu^{2+} and Zn^{2+}) and also for Rural Sites 1 and 2, located 50 km and 100 km east, respectively, of the urban site (see Figure 4).

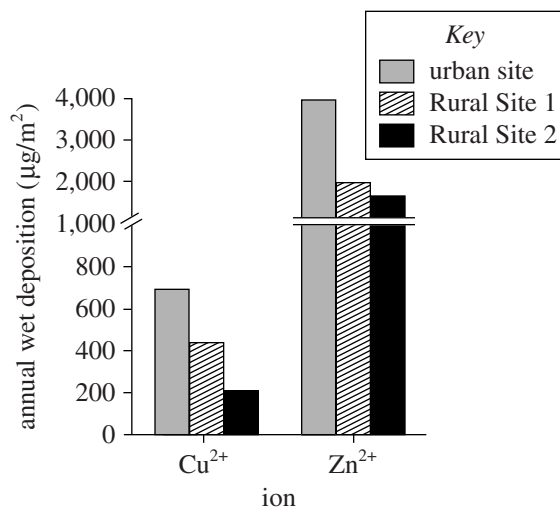


Figure 4

Figures adapted from Kathryn Conko et al., "Atmospheric Wet Deposition of Trace Elements to a Suburban Environment, Reston, Virginia, USA." ©2004 by Elsevier, Ltd.



7. According to Figure 1, over the 12-month period, the monthly precipitation at the urban site was maximum in February and minimum in July. According to Figures 2 and 3, the wet deposition of which ion was also maximum in February and minimum in July?
- A. Cu^{2+}
 - B. Zn^{2+}
 - C. Cl^-
 - D. SO_4^{2-}
8. Based on the results of Study 1, the average monthly wet deposition for Cu^{2+} over the 12-month period was:
- F. less than $50 \mu\text{g}/\text{m}^2$.
 - G. between $50 \mu\text{g}/\text{m}^2$ and $75 \mu\text{g}/\text{m}^2$.
 - H. between $75 \mu\text{g}/\text{m}^2$ and $100 \mu\text{g}/\text{m}^2$.
 - J. greater than $100 \mu\text{g}/\text{m}^2$.
9. Is the statement “The values for Cl^- wet deposition were greater during the winter and early spring when road salt is typically applied” supported by the results of Study 2 ?
- A. Yes, because Cl^- wet deposition values were, on average, greater from November to April than they were from May to October.
 - B. Yes, because Cl^- wet deposition values were, on average, less from November to April than they were from May to October.
 - C. No, because Cl^- wet deposition values were, on average, greater from November to April than they were from May to October.
 - D. No, because Cl^- wet deposition values were, on average, less from November to April than they were from May to October.
10. Suppose there had been no precipitation during 1 entire month of the 12-month period. Based on the information provided, during that month there would have been:
- F. significant wet deposition of all 4 substances.
 - G. significant wet deposition of Cu^{2+} and Zn^{2+} , but no wet deposition of Cl^- and SO_4^{2-} .
 - H. no wet deposition of any of the 4 substances.
 - J. no wet deposition of Cu^{2+} and Zn^{2+} , but significant wet deposition of Cl^- and SO_4^{2-} .
11. According to Study 3, as distance from the urban site increased, the annual wet deposition:
- A. increased for both Cu^{2+} and Zn^{2+} .
 - B. increased for Cu^{2+} but decreased for Zn^{2+} .
 - C. decreased for both Cu^{2+} and Zn^{2+} .
 - D. remained the same for both Cu^{2+} and Zn^{2+} .
12. Which of the following variables was kept constant in Study 2 ?
- F. Site
 - G. Monthly rainfall
 - H. Wet deposition of Zn^{2+}
 - J. Wet deposition of Cl^-



Passage III

Cloud cover is the percent of Earth's surface covered by clouds. Cloud cover may increase because of an increase in the *cosmic ray flux* (number of high-energy particles from space reaching Earth per m^2 per hour). Table 1 shows how Earth's cover of *low clouds* (0 km to 3.2 km altitude) varies with the cosmic ray flux. Figures 1–3 show the *relative cosmic ray flux*, RCRF (the percent below the flux measured on October 1, 1965), and the monthly average cover of *high clouds* (6.0 km to 16.0 km altitude), *middle clouds* (3.2 km to 6.0 km altitude), and low clouds, respectively, from January 1980 to January 1995.

Table 1	
Cosmic ray flux (particles/ m^2/hr)	Cover of low clouds (%)
340,000	27.8
360,000	28.1
380,000	28.4
400,000	28.7
420,000	29.0

Table 1 adapted from E. Palle Bagó and C. J. Butler, "The Influence of Cosmic Rays on Terrestrial Clouds and Global Warming." ©2000 by Institute of Physics Publications, Ltd.

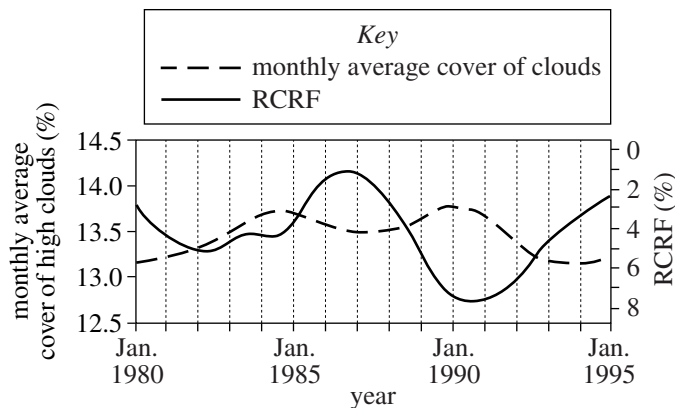


Figure 1

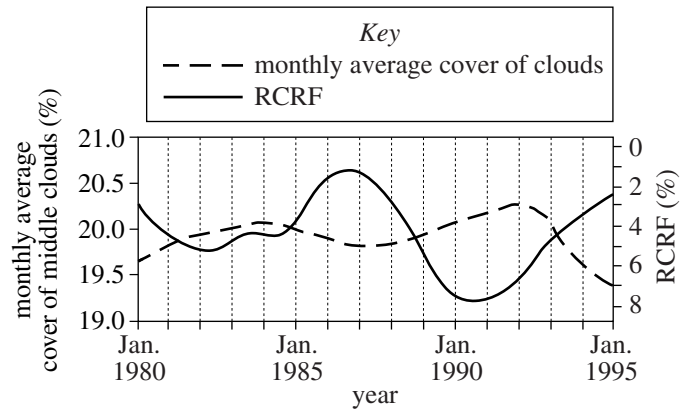


Figure 2

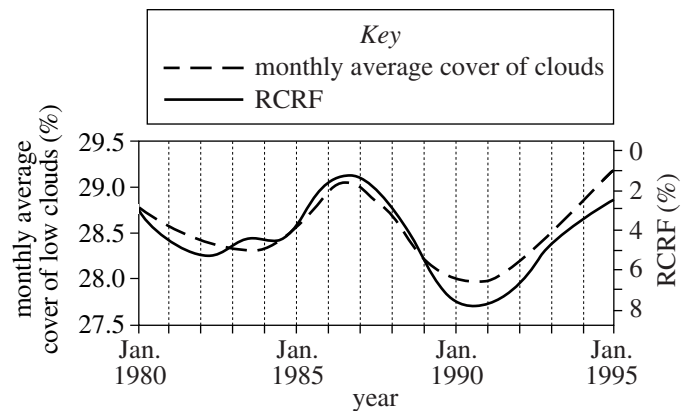
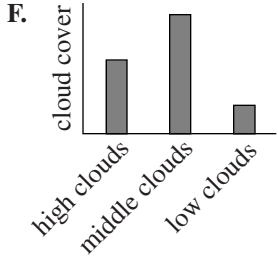
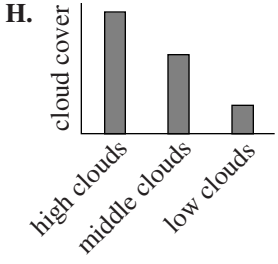
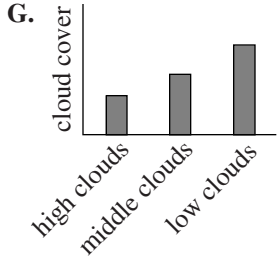
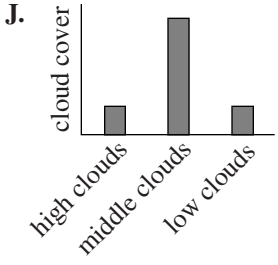


Figure 3

Figures adapted from Nigel Marsh and Henrik Svensmark, "Low Cloud Properties Influenced by Cosmic Rays." ©2000 by The American Physical Society.



13. The percent of Earth's surface covered by high clouds in January 1987 was closest to which of the following?
- 13.0%
 - 13.5%
 - 14.0%
 - 14.5%
14. Based on Table 1, a cosmic ray flux of 440,000 particles/m²/hr would correspond to a cover of low clouds that is closest to which of the following?
- 28.7%
 - 29.0%
 - 29.3%
 - 29.6%
15. Is the statement "The monthly average cover of low clouds is more directly correlated with cosmic ray flux than is the monthly average cover of high clouds" consistent with Figures 1 and 3 ?
- Yes, because the plot for the monthly average cover of low clouds more closely parallels the plot for RCRF.
 - Yes, because the plot for the monthly average cover of high clouds more closely parallels the plot for RCRF.
 - No, because the plot for the monthly average cover of low clouds more closely parallels the plot for RCRF.
 - No, because the plot for the monthly average cover of high clouds more closely parallels the plot for RCRF.
16. Which of the following figures best represents the monthly average cover of high, middle, and low clouds in January 1992 ?
- F.** 
- H.** 
- G.** 
- J.** 
17. High clouds are composed primarily of ice crystals, whereas low clouds are composed primarily of water droplets. This difference is most likely because the average air temperature at altitudes from:
- 0 km to 3.2 km is at or below 0°C, whereas the average air temperature at altitudes from 3.2 km to 6.0 km is above 0°C.
 - 0 km to 3.2 km is at or below 0°C, whereas the average air temperature at altitudes from 6.0 km to 16.0 km is above 0°C.
 - 0 km to 3.2 km is above 0°C, whereas the average air temperature at altitudes from 3.2 km to 6.0 km is at or below 0°C.
 - 0 km to 3.2 km is above 0°C, whereas the average air temperature at altitudes from 6.0 km to 16.0 km is at or below 0°C.



Passage IV

Acid-base titration is a technique in which precise volumes of a *titrant* (an acid or base solution) are added incrementally to a known volume of a *sample solution* (a base or acid solution, respectively). This process can be monitored by adding an *acid-base indicator* (a substance that changes color over a certain pH range) to the sample solution or by measuring the sample solution's *conductivity*. Conductivity (measured in kilosiemens per centimeter, kS/cm) is a measure of a substance's ability to conduct electricity.

Two titration experiments were done at 25°C using a 0.10 M sodium hydroxide (NaOH) solution and either a 0.0010 M hydrochloric acid (HCl) solution or a 0.0010 M acetic acid solution (where M is moles of acid or base per liter of solution). All solutions were aqueous. An acid-base indicator solution of *nitrazine yellow* was also used. Nitrazine yellow is yellow if the pH is less than 6.0 or blue if the pH is greater than 7.0.

Experiment 2

Experiment 1 was repeated, except that the acetic acid solution was used instead of the HCl solution (see Figure 2).

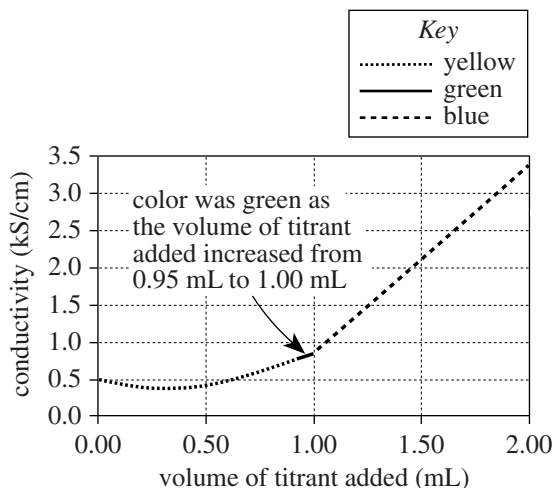


Figure 2

Figures adapted from J. West Loveland, "Conductance and Oscillography," in Gary D. Christian and James E. O'Reilly, eds., *Instrumental Analysis*, 2nd ed. ©1986 by Allyn and Bacon, Inc.

Experiment 1

A drop of nitrazine yellow solution was added to a flask containing 100.0 mL of the HCl solution. A probe that measures conductivity was placed in the solution. The NaOH solution was slowly added to the HCl solution in small increments. After each addition, the HCl solution was stirred and then the solution's color and conductivity were recorded (see Figure 1).

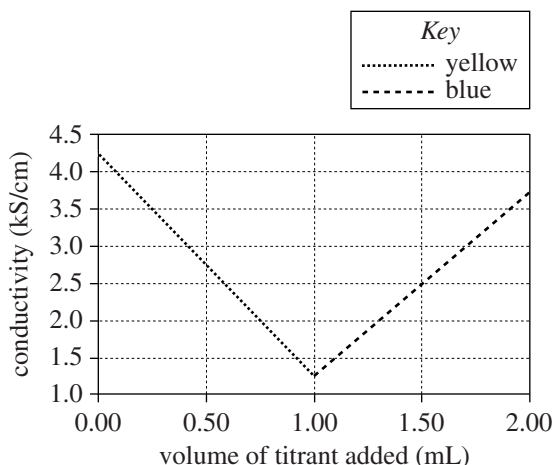


Figure 1

18. In Experiment 1, the sample solution was yellow at which of the following values for the volume of titrant added?

F. 0.80 mL
G. 1.20 mL
H. 1.60 mL
J. 2.00 mL

19. In Experiment 2, the sample solution was neutral at which of the following values for the volume of titrant added?

A. 0.50 mL
B. 1.00 mL
C. 1.50 mL
D. 2.00 mL



20. In Experiment 1, if 2.30 mL of titrant had been added to the sample solution, the conductivity would most likely have been:
- F. less than 0.80 kS/cm.
 - G. between 0.80 kS/cm and 2.30 kS/cm.
 - H. between 2.30 kS/cm and 3.80 kS/cm.
 - J. greater than 3.80 kS/cm.
21. In Experiment 2, which solution was the titrant and which solution was the sample solution?
- | | <u>titrant</u> | <u>sample solution</u> |
|----|----------------|------------------------|
| A. | acetic acid | NaOH |
| B. | HCl | NaOH |
| C. | NaOH | acetic acid |
| D. | NaOH | HCl |
22. In Experiments 1 and 2, the probe that was placed in the sample solution most likely did which of the following?
- F. Cooled the solution to its freezing point
 - G. Heated the solution to its boiling point
 - H. Detected the concentration of nitrazine yellow in the solution
 - J. Passed an electrical current through a portion of the solution
23. A chemist claimed that in Experiment 2, the pH of the sample solution was greater at a value of 0.2 mL of titrant added than at a value of 1.8 mL of titrant added. Do the results of Experiment 2 support this claim?
- A. No; at a value of 0.2 mL of titrant added, the sample solution was yellow, and at a value of 1.8 mL of titrant added, the sample solution was blue.
 - B. No; at a value of 0.2 mL of titrant added, the sample solution was blue, and at a value of 1.8 mL of titrant added, the sample solution was yellow.
 - C. Yes; at a value of 0.2 mL of titrant added, the sample solution was yellow, and at a value of 1.8 mL of titrant added, the sample solution was blue.
 - D. Yes; at a value of 0.2 mL of titrant added, the sample solution was blue, and at a value of 1.8 mL of titrant added, the sample solution was yellow.

**Passage V**

An astronomy class is given the following facts about stellar evolution.

1. A star's evolution can be divided into 3 stages: *pre-main sequence* (pre-MS), *main sequence* (MS), and *post-main sequence* (post-MS).
2. Gravity causes part of a cloud of gas and dust to collapse and heat up, creating a pre-MS star. The star's hot dust and gas emit its energy.
3. A pre-MS star becomes an MS star when the star produces the majority of its energy by fusing hydrogen nuclei (protons) at its center to make helium nuclei.
4. An MS star becomes a post-MS star when the star expands in volume and produces the majority of its energy by fusing hydrogen to make helium in a shell surrounding its center.
5. The more massive a star, the more rapidly the star passes through each of the 3 stages of its evolution.

Two students discuss the evolution of the *Algol system*—Algol A, a 3.6-solar-mass MS star; Algol B, a 0.8-solar-mass post-MS star; and Algol C, a 1.7-solar-mass MS star. (One solar mass = the Sun's mass.) The 3 stars orbit a mutual center of mass, with Algol A and Algol B much closer to each other and to the center of mass than to Algol C.

Student 1

The 3 stars of the Algol system formed at the same time from the same cloud of gas and dust. Algol B, originally the most massive of the 3 stars, became a post-MS star and expanded in volume while Algol A remained an MS star. Because the matter in the outer parts of Algol B was more strongly attracted to Algol A than to the matter in the inner parts of Algol B, this matter flowed from Algol B to Algol A, and, over time, Algol A became more massive than Algol B.

Student 2

Algol B was not part of the original Algol system (Algol A and Algol C). Algol B and the original Algol system formed in different clouds of gas and dust at different times and moved in 2 different but intersecting orbits around the center of the galaxy. During a particular orbit, Algol B encountered the original Algol system at the intersection of the 2 orbits and became part of the Algol system.

Algol B became a post-MS star while Algol A and Algol C remained MS stars. Algol B never lost mass to Algol A. Algol B was always less massive than Algol A.

24. Based on Student 2's discussion, Algol B is part of the present Algol system because of which of the following forces exerted on Algol B by the original Algol system?

F. Electric force
G. Magnetic force
H. Gravitational force
J. Nuclear force

25. Based on Student 1's discussion and Fact 4, while matter flowed between Algol A and Algol B, Algol B produced the majority of its energy by fusing:

A. hydrogen nuclei to make helium nuclei at its center.
B. hydrogen nuclei to make helium nuclei in a shell surrounding its center.
C. helium nuclei to make hydrogen nuclei at its center.
D. helium nuclei to make hydrogen nuclei in a shell surrounding its center.

26. Suppose that chemical composition is uniform among stars formed from the same cloud of gas and dust, but that chemical composition varies among stars formed from different clouds of gas and dust. Student 2 would most likely agree with which of the following statements comparing the chemical compositions of the stars in the present-day Algol system at the time they formed?

F. Algol A and Algol B had the most similar compositions.
G. Algol A and Algol C had the most similar compositions.
H. Algol B and Algol C had the most similar compositions.
J. Algol A, Algol B, and Algol C had the same composition.

27. If the mass of the Sun is 2.0×10^{30} kg, what is the mass of Algol C?

A. 1.6×10^{30} kg
B. 2.0×10^{30} kg
C. 3.4×10^{30} kg
D. 7.2×10^{30} kg

28. Which of the following statements best explains why the reaction described in Fact 3 requires a high temperature and pressure?

F. All protons are positively charged, and like charges attract each other.
G. All protons are positively charged, and like charges repel each other.
H. All electrons are negatively charged, and like charges attract each other.
J. All electrons are negatively charged, and like charges repel each other.



29. Based on Fact 5 and Student 1's discussion, which of the 3 stars in the Algol system, if any, was most likely the first to become an MS star?

- A.** Algol A
- B.** Algol B
- C.** Algol C
- D.** The 3 stars became MS stars at the same time.

30. Based on Fact 5, would Student 2 agree that by the time Algol A stops being an MS star, Algol A will have spent as much time being an MS star as Algol B spent being an MS star?

- F.** Yes, because according to Student 2, Algol A has always been more massive than Algol B.
- G.** Yes, because according to Student 2, Algol A has always been less massive than Algol B.
- H.** No, because according to Student 2, Algol A has always been more massive than Algol B.
- J.** No, because according to Student 2, Algol A has always been less massive than Algol B.

**Passage VI**

Three experiments were done using CO₂, krypton (Kr), or O₂. For each gas:

1. A 3 L steel vessel was fitted with a cap that contained a gas inlet valve and a pressure and temperature sensor.
2. Air was pumped out of the vessel until the pressure measured 0.00 torr.
3. The vessel was placed on a balance, and the balance was reset to 0.000 g.
4. Some of the gas was added to the vessel.
5. When the gas in the vessel reached room temperature (22°C), mass and pressure were recorded.
6. Steps 4 and 5 were repeated several times.

The experiments were then repeated, except that a 6 L vessel was used (see Figures 1 and 2).

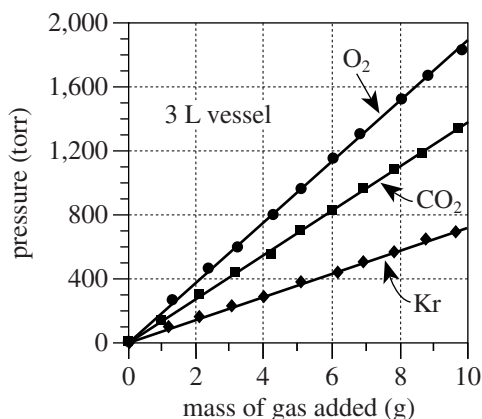


Figure 1

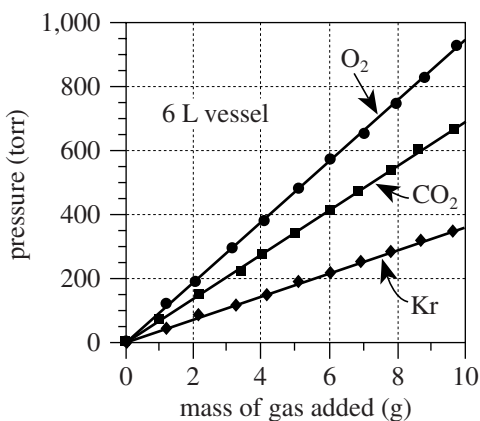


Figure 2

31. Based on Figure 2, if 13 g of Kr had been added to the 6 L vessel, the pressure would have been:
 - A. less than 200 torr.
 - B. between 200 torr and 400 torr.
 - C. between 400 torr and 600 torr.
 - D. greater than 600 torr.
32. Suppose the experiments had been repeated, except with a 5 L vessel. Based on Figures 1 and 2, the pressure exerted by 7 g of CO₂ would most likely have been:
 - F. less than 500 torr.
 - G. between 500 torr and 1,000 torr.
 - H. between 1,000 torr and 1,500 torr.
 - J. greater than 1,500 torr.
33. Based on Figures 1 and 2, for a given mass of O₂ at 22°C, how does the pressure exerted by the O₂ in a 6 L vessel compare to the pressure exerted by the O₂ in a 3 L vessel? In the 6 L vessel, the O₂ pressure will be:
 - A. $\frac{1}{2}$ as great as in the 3 L vessel.
 - B. the same as in the 3 L vessel.
 - C. 2 times as great as in the 3 L vessel.
 - D. 4 times as great as in the 3 L vessel.
34. Which of the following best explains why equal masses of O₂ and CO₂ at the same temperature and in the same-size vessel had different pressures? The pressure exerted by the O₂ was:
 - F. less, because there were fewer O₂ molecules per gram than there were CO₂ molecules per gram.
 - G. less, because there were more O₂ molecules per gram than there were CO₂ molecules per gram.
 - H. greater, because there were fewer O₂ molecules per gram than there were CO₂ molecules per gram.
 - J. greater, because there were more O₂ molecules per gram than there were CO₂ molecules per gram.
35. Suppose the experiment involving O₂ and the 6 L vessel had been repeated, except at a room temperature of 14°C. For a given mass of O₂, compared to the pressure measured in the original experiment, the pressure measured at 14°C would have been:
 - A. less, because pressure is directly proportional to temperature.
 - B. less, because pressure is inversely proportional to temperature.
 - C. greater, because pressure is directly proportional to temperature.
 - D. greater, because pressure is inversely proportional to temperature.



Passage VII

The *human threshold of hearing* is the minimum intensity at each sound frequency required for a sound to be heard by humans. The *human threshold of pain* is the maximum intensity at each sound frequency that humans can tolerate without pain.

The figure below displays, for sounds in water and in air, the human thresholds of hearing and of pain. The figure also shows S , the percent increase in air density and water density that accompanies the compression of air and water by sound waves of given intensities. Sound intensities are given in decibels (db) and frequencies are given in hertz [(Hz); 1 Hz = 1 cycle/sec].

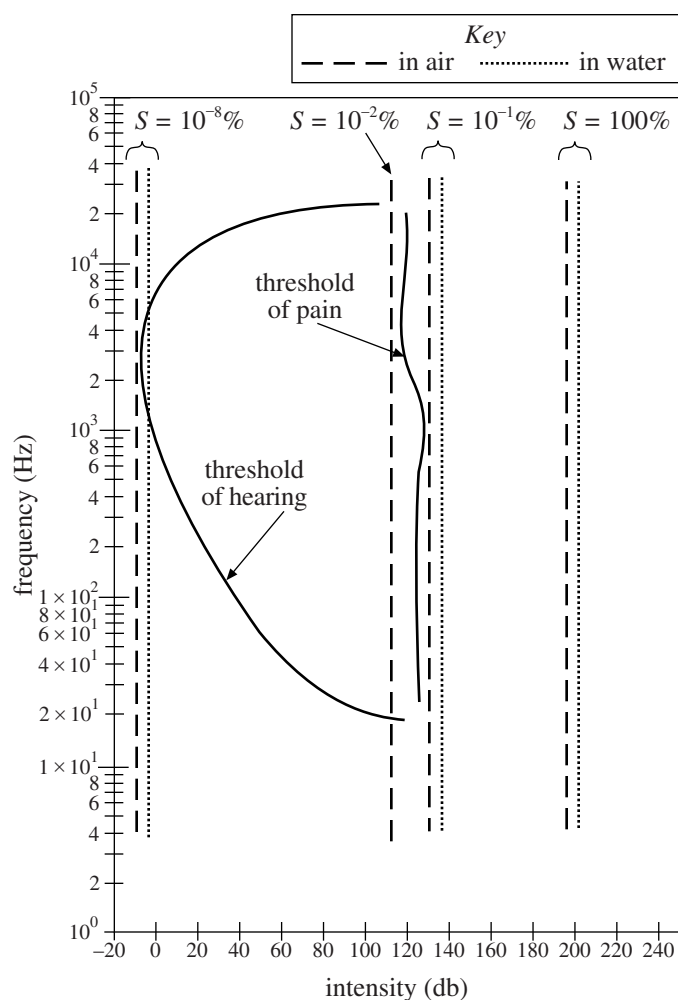
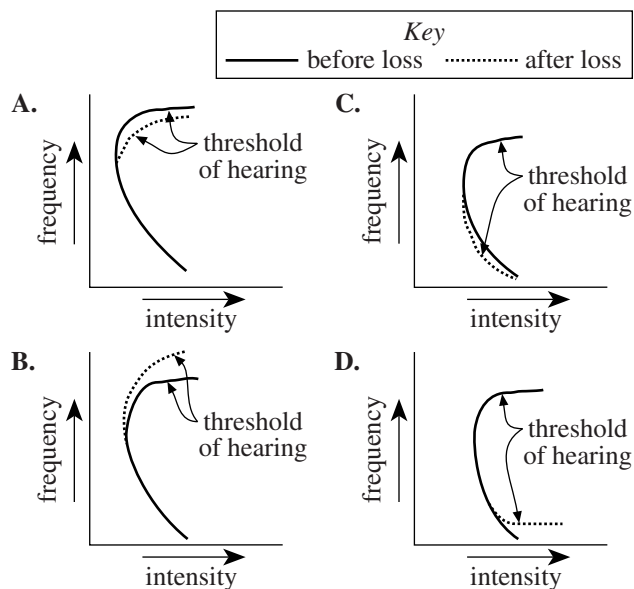


Figure adapted from Rita G. Lerner and George L. Trigg, eds., *Encyclopedia of Physics*, 2nd ed. ©1991 by VCH Publishers, Inc.

36. According to the figure, which of the following is closest to the lowest frequency that can be heard by a human being?

- F. 8 Hz
- G. 20 Hz
- H. 1,000 Hz
- J. 20,000 Hz

37. As humans age, it is common for selective hearing loss to occur at high sound frequencies. Which of the following figures best illustrates this loss?



38. Based on the figure, a sound of a given frequency will have the highest intensity for which of the following sets of conditions?

Sound is passing through:		S
F.	water	100%
G.	water	$10^{-8}\%$
H.	air	100%
J.	air	$10^{-8}\%$

39. A student hypothesized that sounds of any intensity at a frequency of 10^5 Hz would be painful for humans to hear. Do the data in the figure support this hypothesis?

- A. Yes, because the threshold of pain is relatively constant with changes in frequency.
- B. Yes, because as frequency increases above 10^5 Hz, the threshold of pain increases.
- C. No, because humans cannot hear sounds at 10^5 Hz.
- D. No, because the threshold of pain is relatively constant with changes in frequency.

40. Based on the figure, does S depend on the frequency of a sound wave of a given intensity?

- F. Yes, because as frequency increases, S increases.
- G. Yes, because as frequency increases, S remains constant.
- H. No, because as frequency increases, S increases.
- J. No, because as frequency increases, S remains constant.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

If you plan to take the ACT Plus Writing, sharpen your pencils and continue with the Writing Test on page 55.

If you do not plan to take the ACT Plus Writing, skip to page 57 for instructions on scoring your multiple-choice tests.

Practice Writing Test

Your Signature: _____
(Do not print.)

Print Your Name Here: _____

Your Date of Birth:									
		-			-				
Month		Day		Year					

Form 14R

The ACT[®] WRITING TEST BOOKLET

You must take the multiple-choice tests before you take the Writing Test.

Directions

This is a test of your writing skills. You will have thirty (30) minutes to write an essay in English. Before you begin planning and writing your essay, read the writing prompt carefully to understand exactly what you are being asked to do. Your essay will be evaluated on the evidence it provides of your ability to express judgments by taking a position on the issue in the writing prompt; to maintain a focus on the topic throughout the essay; to develop a position by using logical reasoning and by supporting your ideas; to organize ideas in a logical way; and to use language clearly and effectively according to the conventions of standard written English.

You may use the unlined pages in this test booklet to plan your essay. These pages will not be scored. ***You must write your essay in pencil on the lined pages in the answer folder.*** Your writing on those lined pages will be scored. You may not need all the lined pages, but to ensure you have enough room to finish, do NOT skip lines. You may write corrections or additions neatly between the lines of your essay, but do NOT write in the margins of the lined pages. ***Illegible essays cannot be scored, so you must write (or print) clearly.***

If you finish before time is called, you may review your work. Lay your pencil down immediately when time is called.

DO NOT OPEN THIS BOOKLET UNTIL TOLD TO DO SO.

ACT Writing Test Prompt

Rather than concentrating on doing one thing at a time, high school students often divide their attention among several activities, such as watching television and using the computer while doing homework. Educators debate whether performing several tasks at the same time is too distracting when students are doing homework. Some educators believe multitasking is a bad practice when doing homework because they think dividing attention between multiple tasks negatively affects the quality of students' work. Other educators do not believe multitasking is a bad practice when doing homework because they think students accomplish more during their limited free time as a result of multitasking. In your opinion, is it too distracting for high school students to divide their attention among several activities when they are doing homework?

In your essay, take a position on this question. You may write about either one of the two points of view given, or you may present a different point of view on this question. Use specific reasons and examples to support your position.

Note

- Your test booklet will have blank space for you to plan your essay. For this practice test, use scratch paper.
- You may wish to remove pages 75–78 to respond to this prompt.
- When you have completed your essay, read pages 64–72 for information and instructions on scoring your practice Writing Test.

5 Scoring Your Tests

How to Score the Multiple-Choice Tests

Follow the instructions below and on the following pages to score your practice multiple-choice tests and to evaluate your performance.

Raw Scores

The number of questions you answered correctly on each test and in each subscore area is your raw score. Because there are many forms of the ACT, each containing different questions, some forms will be slightly easier (and some slightly harder) than others. A raw score of 67 on one form of the English Test, for example, may be about as difficult to earn as a raw score of 70 on another form of that test.

To compute your raw scores, check your answers with the scoring keys on pages 58–60. Count the number of correct answers for each of the four tests and seven subscore areas, and enter the number in the blanks provided on those pages. These numbers are your raw scores on the tests and subscore areas.

Scale Scores

To adjust for the small differences that occur among different forms of the ACT, the raw scores for tests and subscore areas are converted into *scale scores*. Scale scores are printed on the reports sent to you and your college and scholarship choices.

When your raw scores are converted into scale scores, it becomes possible to compare your scores with those of examinees who took different test forms. For example, a scale score of 26 on the English Test has the same meaning regardless of the form of the ACT on which it is based.

To determine the scale scores corresponding to your raw scores on the practice test, use the score conversion tables on pages 61–62. Table 1 on page 61 shows the raw-to-scale score conversions for each test, and Table 2 on page 62 shows the raw-to-scale score conversions for the subscore areas. Because each form of the ACT is unique, each form has somewhat different conversion tables. Consequently, these tables provide only approximations of the raw-to-scale score conversions that would apply if a different form of the ACT were taken. Therefore, the scale scores obtained from the practice tests don't match precisely the scale scores received from an actual administration of the ACT.

Computing the Composite Score

The Composite score is the average of the four scale scores in English, Mathematics, Reading, and Science. If you left any of these tests blank, do not calculate a Composite score. If you take the ACT Plus Writing, your Writing results do **not** affect your Composite score.

Comparing Your Scores

Even scale scores don't tell the whole story of your test performance. You may want to know how your scores compare to the scores of other students who took the ACT.

The multiple-choice norms table (Table 3A on page 63) enables you to compare your scores on the practice multiple-choice tests with the scores of recent high school graduates who took the ACT. The numbers reported in Table 3A are cumulative percents. A cumulative percent is the percent of students who scored *at or below* a given score. For example, a Composite score of 20 has a cumulative percent of 48. This means that 48% of the ACT-tested high school students had a Composite score of 20 or lower.

Remember that your scores and percent at or below on the practice test are only *estimates* of the scores that you will obtain during an actual administration of the ACT. Test scores are only one indicator of your level of academic knowledge and skills. Consider your scores in connection with your grades, your performance in outside activities, and your career interests.

College Readiness Standards™

To add to the information you receive about your performance on the ACT, we have developed College Readiness Standards. These standards help you to more fully understand what your total test score means for each academic area assessed: English, Mathematics, Reading, Science, and Writing. The College Readiness Standards describe the types of skills, strategies, and understandings you will need to make a successful transition from high school to college. For English, Mathematics, Reading, and Science, standards are provided for six score ranges that reflect the progression and complexity of the skills in each of the academic areas measured by the ACT tests. For Writing, standards are provided for five score ranges. The College Readiness Standards and benchmark scores for each test can be found at www.act.org/standard and www.act.org/education/benchmarks.html.

Reviewing Your Performance on the Practice Multiple-Choice Tests

After you have determined your scale scores, consider the following as you evaluate your performance.

- Did you run out of time? If so, reread the information in this booklet on pacing yourself. Perhaps you need to adjust the way you used your time in responding to the questions. It is to your advantage to answer every question. There is no penalty for guessing.
- Did you spend too much time trying to understand the directions for the tests? The directions for the practice tests are the same directions that will appear in your test booklet on test day. Make sure you understand them now, so you won't have to spend too much time studying them on test day.
- Review the questions that you missed. Did you select a response that was an incomplete answer or that did not directly respond to the question being asked? Try to figure out what you overlooked in answering the questions.
- Did a particular type of question confuse you? Did the questions you missed come from a particular subscore area? In reviewing your responses, check to see whether a particular type of question or a particular subscore area was more difficult for you or took more time.

Scoring Keys for the ACT Practice Tests

Use the scoring key for each test to score your answer document for the multiple-choice tests. Mark a “1” in the blank for each question you answered correctly. Add up the numbers in each subscore area and enter the total number correct for each subscore area in the blanks provided. Also enter the total number correct for each test in the blanks provided. The total number correct for each test is the sum of the number correct in each subscore area.

Test 1: English—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	UM	RH	Key	UM	RH	Key	UM	RH
1. B	_____		26. G	_____		51. B		_____
2. J	_____		27. A		_____	52. J		_____
3. D		_____	28. G		_____	53. D	_____	
4. F	_____		29. C		_____	54. H		_____
5. B		_____	30. J		_____	55. A	_____	
6. J	_____		31. B		_____	56. H	_____	
7. D		_____	32. J		_____	57. B		_____
8. F	_____		33. D		_____	58. H	_____	
9. A		_____	34. H	_____		59. A		_____
10. H	_____		35. B	_____		60. G		_____
11. A		_____	36. H	_____		61. A	_____	
12. H	_____		37. C	_____		62. H	_____	
13. D	_____		38. F	_____		63. A	_____	
14. G	_____		39. C		_____	64. H		_____
15. B		_____	40. J		_____	65. D	_____	
16. G	_____		41. B	_____		66. G	_____	
17. C	_____		42. F		_____	67. A	_____	
18. G		_____	43. B	_____		68. J	_____	
19. D		_____	44. J		_____	69. A	_____	
20. F		_____	45. A	_____		70. G	_____	
21. B	_____		46. J	_____		71. D	_____	
22. J	_____		47. C	_____		72. J	_____	
23. C	_____		48. F		_____	73. C	_____	
24. F		_____	49. B	_____		74. G	_____	
25. C	_____		50. F	_____		75. D		_____

Number Correct (Raw Score) for:

Usage/Mechanics (UM) Subscore Area	_____
	(40)
Rhetorical Skills (RH) Subscore Area	_____
	(35)
Total Number Correct for English Test (UM + RH)	_____
	(75)

*UM = Usage/Mechanics
RH = Rhetorical Skills

1267C

Test 2: Mathematics—Scoring Key

Subscore Area*					Subscore Area*				
Key	EA	AG	GT		Key	EA	AG	GT	
1. A	_____				31. E			_____	
2. F	_____				32. G	_____			
3. E	_____				33. D		_____		
4. J	_____				34. F		_____		
5. C	_____				35. D	_____			
6. K	_____				36. F		_____		
7. E	_____				37. B			_____	
8. H	_____				38. G			_____	
9. A	_____				39. C		_____		
10. J			_____		40. G			_____	
11. D			_____		41. B		_____		
12. K	_____				42. F			_____	
13. B	_____				43. D	_____			
14. F		_____			44. H			_____	
15. C		_____			45. A			_____	
16. J	_____				46. J	_____			
17. B			_____		47. C			_____	
18. J			_____		48. H			_____	
19. D	_____				49. B		_____		
20. J			_____		50. F		_____		
21. E	_____				51. C	_____			
22. G			_____		52. J		_____		
23. B	_____				53. E		_____		
24. H		_____			54. K	_____			
25. A		_____			55. C	_____			
26. G	_____				56. J			_____	
27. B			_____		57. E			_____	
28. H		_____			58. G		_____		
29. E		_____			59. C	_____			
30. K		_____			60. F		_____		

Number Correct (Raw Score) for:	
Pre-Alg./Elem. Alg. (EA) Subscore Area	_____ (24)
Inter. Alg./Coord. Geo. (AG) Subscore Area	_____ (18)
Plane Geo./Trig. (GT) Subscore Area	_____ (18)
Total Number Correct for Math Test (EA + AG + GT)	_____ (60)

*EA = Pre-Algebra/Elementary Algebra
AG = Intermediate Algebra/Coordinate Geometry
GT = Plane Geometry/Trigonometry

1267C

Test 3: Reading—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	SS	AL	Key	SS	AL	Key	SS	AL
1. D		_____	15. D	_____		29. A		_____
2. H		_____	16. H	_____		30. F		_____
3. B		_____	17. B	_____		31. B	_____	
4. G		_____	18. J	_____		32. H	_____	
5. B		_____	19. A	_____		33. D	_____	
6. F		_____	20. F	_____		34. J	_____	
7. C		_____	21. C		_____	35. B	_____	
8. F		_____	22. J		_____	36. G	_____	
9. D		_____	23. A		_____	37. B	_____	
10. H		_____	24. J		_____	38. F	_____	
11. C	_____		25. C		_____	39. A	_____	
12. F	_____		26. F		_____	40. F	_____	
13. C	_____		27. C		_____			
14. G	_____		28. J		_____			

Number Correct (Raw Score) for:

Social Studies/Sciences (SS) Subscore Area	_____
	(20)
Arts/Literature (AL) Subscore Area	_____
	(20)
Total Number Correct for Reading Test (SS + AL)	_____
	(40)

*SS = Social Studies/Sciences
AL = Arts/Literature

Test 4: Science—Scoring Key

Key	Key	Key
1. D	15. A	29. B
2. J	16. G	30. H
3. B	17. D	31. C
4. J	18. F	32. G
5. C	19. B	33. A
6. F	20. J	34. J
7. D	21. C	35. A
8. G	22. J	36. G
9. A	23. A	37. A
10. H	24. H	38. F
11. C	25. B	39. C
12. F	26. G	40. J
13. B	27. C	
14. H	28. G	

Number Correct (Raw Score) for:

Total Number Correct for Science Test	_____
	(40)

TABLE 1**Explanation of Procedures Used to Obtain Scale Scores from Raw Scores**

On each of the four multiple-choice tests on which you marked any responses, the total number of correct responses yields a raw score. Use the table below to convert your raw scores to scale scores. For each test, locate and circle your raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale score that corresponds to that raw score. As you determine your scale scores, enter them in the blanks provided on the right. The highest possible scale score for each test is 36. The lowest possible scale score for any test on which you marked any responses is 1.

Next, compute the Composite score by averaging the four scale scores. To do this, add your four scale scores and divide the sum by 4. If the resulting number ends in a fraction, round it off to the nearest whole number. (Round down any fraction less than one-half; round up any fraction that is one-half or more.) Enter this number in the blank. This is your Composite score. The highest possible Composite score is 36. The lowest possible Composite score is 1.

ACT Test 67C**Your Scale Score**

English _____

Mathematics _____

Reading _____

Science _____

Sum of scores _____**Composite score (sum ÷ 4)** _____

NOTE: If you left a test completely blank and marked no items, do not list a scale score for that test. If any test was completely blank, do not calculate a Composite score.

Scale Score	Raw Scores				Scale Score
	Test 1 English	Test 2 Mathematics	Test 3 Reading	Test 4 Science	
36	75	59-60	40	40	36
35	73-74	57-58	39	39	35
34	71-72	55-56	38	38	34
33	70	54	—	37	33
32	69	53	37	—	32
31	68	52	36	36	31
30	67	50-51	35	35	30
29	66	49	34	34	29
28	64-65	47-48	33	33	28
27	62-63	45-46	32	31-32	27
26	60-61	43-44	31	30	26
25	58-59	41-42	30	28-29	25
24	56-57	38-40	29	26-27	24
23	53-55	36-37	27-28	24-25	23
22	51-52	34-35	26	23	22
21	48-50	33	25	21-22	21
20	45-47	31-32	23-24	19-20	20
19	42-44	29-30	22	17-18	19
18	40-41	27-28	20-21	16	18
17	38-39	24-26	19	14-15	17
16	35-37	19-23	18	13	16
15	33-34	15-18	16-17	12	15
14	30-32	12-14	14-15	11	14
13	29	10-11	13	10	13
12	27-28	8-9	11-12	9	12
11	25-26	6-7	9-10	8	11
10	23-24	5	8	7	10
9	20-22	4	7	6	9
8	17-19	—	6	5	8
7	14-16	3	5	4	7
6	11-13	—	4	3	6
5	9-10	2	3	—	5
4	6-8	—	—	2	4
3	5	1	2	1	3
2	3-4	—	1	—	2
1	0-2	0	0	0	1

TABLE 2

Explanation of Procedures Used to Obtain Scale Subscores from Raw Scores

For each of the seven subscore areas, the total number of correct responses yields a raw score. Use the table below to convert your raw scores to scale subscores. For each of the seven subscore areas, locate and circle either the raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale subscore that corresponds to that raw score. As you determine your scale subscores, enter them in the blanks provided on the right. The highest possible scale subscore is 18. The lowest possible scale subscore is 1.

If you left a test completely blank and marked no responses, do not list any scale subscores for that test.

ACT Test 67C Your Scale Subscore

English

Usage/Mechanics _____

Rhetorical Skills _____

Mathematics

Pre-Algebra/Elementary Algebra _____

Intermed. Algebra/Coord. Geometry _____

Plane Geometry/Trigonometry _____

Reading

Social Studies/Sciences _____

Arts/Literature _____

Scale Subscore	Raw Scores							Scale Subscore
	Test 1 English		Test 2 Mathematics			Test 3 Reading		
	Usage/ Mechanics	Rhetorical Skills	Pre-Algebra/ Elem. Algebra	Inter. Algebra/ Coord. Geometry	Plane Geometry/ Trigonometry	Social Studies/ Sciences	Arts/ Literature	
18	39-40	35	23-24	18	18	20	20	18
17	37-38	34	22	17	17	19	18-19	17
16	35-36	33	21	16	16	18	—	16
15	34	31-32	20	15	14-15	17	17	15
14	32-33	29-30	19	13-14	13	16	16	14
13	31	27-28	18	12	11-12	15	15	13
12	29-30	25-26	17	10-11	10	14	14	12
11	27-28	22-24	16	9	9	12-13	13	11
10	24-26	20-21	15	7-8	7-8	11	12	10
9	22-23	18-19	13-14	6	6	9-10	11	9
8	20-21	15-17	11-12	4-5	5	8	10	8
7	18-19	13-14	9-10	—	4	6-7	9	7
6	16-17	12	6-8	3	3	5	8	6
5	14-15	10-11	5	2	—	4	6-7	5
4	12-13	8-9	3-4	—	2	3	5	4
3	9-11	5-7	2	1	—	2	3-4	3
2	6-8	3-4	1	—	1	1	2	2
1	0-5	0-2	0	0	0	0	0-1	1

TABLES 3A and 3B

Norms Tables

Use the norms tables below (3A and 3B) to determine your estimated percent at or below for each of your multiple-choice scale scores (3A), and for your Writing scores (3B), if applicable.

In the far left column of the multiple-choice norms table (3A), circle your scale score for the English Test (from page 61). Then read across to the percent at or below column for that test; circle or put a check mark beside the corresponding percent at or below. Use the same procedure for each test and subscore area. Use the far right column of scale scores in Table 3A, for your Science Test and Composite scores. Follow the same procedure on the Writing Test norms to get your estimated percent at or below for your Writing subscore and Combined English/Writing score.

As you mark your percents at or below, enter them in the blanks provided at the right. You may also find it helpful to compare your performance with the national mean (average) score for each of the tests, subscore areas, and the Composite as shown at the bottom of the norms tables.

**Your Estimated
Percent At or Below
on Practice Test**

English

Usage/Mechanics

Rhetorical Skills

Mathematics

Pre-Algebra/Elem. Alg.

Alg./Coord. Geometry

Plane Geometry/Trig.

Reading

Soc. Studies/Sciences

Arts/Literature

Science

Composite

Combined English/Writing

Writing

3A

National Distributions of Cumulative Percents for ACT Test Scores ACT-Tested High School Graduates from 2009, 2010, and 2011

Score	ENGLISH	Usage/Mechanics	Rhetorical Skills	MATHEMATICS	Pre-Algebra/Elem. Alg.	Alg./Coord. Geometry	Plane Geometry/Trig.	READING	Soc. Studies/Sciences	Arts/Literature	SCIENCE	COMPOSITE	Score
36	99			99				99			99	99	36
35	99			99				99			99	99	35
34	99			99				99			99	99	34
33	97			98				97			99	99	33
32	96			97				95			98	98	32
31	94			96				93			98	97	31
30	93			94				91			96	95	30
29	91			93				87			95	93	29
28	88			91				85			93	91	28
27	85			88				82			91	87	27
26	82			84				78			88	84	26
25	78			79				75			84	79	25
24	73			74				71			77	74	24
23	68			67				66			71	68	23
22	63			61				60			63	62	22
21	57			57				54			56	55	21
20	50			52				48			47	48	20
19	43			47				42			38	41	19
18	38	99	99	41	99	99	99	35	99	99	31	34	18
17	33	96	99	34	96	99	99	30	97	97	24	28	17
16	29	92	98	26	91	98	98	25	94	91	19	21	16
15	24	88	92	14	87	95	95	19	88	85	14	16	15
14	18	83	86	06	81	92	90	15	82	77	11	11	14
13	14	78	79	02	74	83	82	10	76	71	08	06	13
12	11	71	71	01	66	74	72	06	69	65	05	03	12
11	09	64	60	01	58	64	63	03	59	57	03	01	11
10	06	56	49	01	48	51	52	01	50	47	02	01	10
09	04	44	38	01	40	36	38	01	40	39	01	01	09
08	02	35	27	01	32	22	25	01	29	31	01	01	08
07	01	27	19	01	20	13	15	01	18	23	01	01	07
06	01	19	12	01	09	07	09	01	10	16	01	01	06
05	01	13	08	01	03	04	06	01	05	09	01	01	05
04	01	07	04	01	01	02	03	01	02	04	01	01	04
03	01	03	02	01	01	01	02	01	01	01	01	01	03
02	01	01	01	01	01	01	01	01	01	01	01	01	02
01	01	01	01	01	01	01	01	01	01	01	01	01	01
Mean	20.6	10.2	10.6	21.0	10.9	10.6	10.5	21.3	10.8	10.9	20.9	21.1	
S.D.	6.4	3.9	3.4	5.3	3.6	2.9	3.1	6.2	3.5	3.9	5.1	5.2	

Note: These norms are the source of national norms, for multiple-choice tests, printed on ACT score reports during the 2011–2012 testing year. Sample size: 4,665,862.

3B

National Distributions of Cumulative Percents for ACT Writing Test Scores

ACT-Tested High School Graduates from 2009, 2010, and 2011

Score	Combined English/Writing	Writing
36	99	
35	99	
34	99	
33	99	
32	99	
31	98	
30	95	
29	93	
28	90	
27	87	
26	84	
25	79	
24	74	
23	69	
22	61	
21	55	
20	47	
19	41	
18	34	
17	28	
16	24	
15	18	
14	14	
13	10	
12	8	99
11	5	99
10	4	99
9	2	94
8	1	86
7	1	51
6	1	37
5	1	11
4	1	7
3	1	2
2	1	1
1	1	
Mean	20.8	7.1
S.D.	5.6	1.6

Note: These norms are the source of the Writing Test norms printed on the ACT score reports of students who take the optional Writing Test during 2011–2012. Sample size: 2,588,762.

Six-Point Holistic Scoring Rubric for the ACT Writing Test

Papers at each level exhibit *all* or *most* of the characteristics described at each score point.

Score = 6

Essays within this score range demonstrate effective skill in responding to the task.

The essay shows a clear understanding of the task. The essay takes a position on the issue and may offer a critical context for discussion. The essay addresses complexity by examining different perspectives on the issue, or by evaluating the implications and/or complications of the issue, or by fully responding to counterarguments to the writer's position. Development of ideas is ample, specific, and logical. Most ideas are fully elaborated. A clear focus on the specific issue in the prompt is maintained. The organization of the essay is clear: the organization may be somewhat predictable or it may grow from the writer's purpose. Ideas are logically sequenced. Most transitions reflect the writer's logic and are usually integrated into the essay. The introduction and conclusion are effective, clear, and well developed. The essay shows a good command of language. Sentences are varied and word choice is varied and precise. There are few, if any, errors to distract the reader.

Score = 5

Essays within this score range demonstrate competent skill in responding to the task.

The essay shows a clear understanding of the task. The essay takes a position on the issue and may offer a broad context for discussion. The essay shows recognition of complexity by partially evaluating the implications and/or complications of the issue, or by responding to counterarguments to the writer's position. Development of ideas is specific and logical. Most ideas are elaborated, with clear movement between general statements and specific reasons, examples, and details. Focus on the specific issue in the prompt is maintained. The organization of the essay is clear, although it may be predictable. Ideas are logically sequenced, although simple and obvious transitions may be used. The introduction and conclusion are clear and generally well developed. Language is competent. Sentences are somewhat varied and word choice is sometimes varied and precise. There may be a few errors, but they are rarely distracting.

Score = 4

Essays within this score range demonstrate adequate skill in responding to the task.

The essay shows an understanding of the task. The essay takes a position on the issue and may offer some context for discussion. The essay may show some recognition of complexity by providing some response to counterarguments to the writer's position. Development of ideas is adequate, with some movement between general statements and specific reasons, examples, and details. Focus on the specific issue in the prompt is maintained throughout most of the essay. The organization of the essay is apparent but predictable. Some evidence of logical sequencing of ideas is apparent, although most transitions are simple and obvious. The introduction and conclusion are clear and somewhat developed. Language is adequate, with some sentence variety and appropriate word choice. There may be some distracting errors, but they do not impede understanding.

Score = 3

Essays within this score range demonstrate some developing skill in responding to the task.

The essay shows some understanding of the task. The essay takes a position on the issue but does not offer a context for discussion. The essay may acknowledge a counterargument to the writer's position, but its development is brief or unclear. Development of ideas is limited and may be repetitious, with little, if any, movement between general statements and specific reasons, examples, and details. Focus on the general topic is maintained, but focus on the specific issue in the prompt may not be maintained. The organization of the essay is simple. Ideas are logically grouped within parts of the essay, but there is little or no evidence of logical sequencing of ideas. Transitions, if used, are simple and obvious. An introduction and conclusion are clearly discernible but underdeveloped. Language shows a basic control. Sentences show a little variety and word choice is appropriate. Errors may be distracting and may occasionally impede understanding.

Score = 2

Essays within this score range demonstrate inconsistent or weak skill in responding to the task.

The essay shows a weak understanding of the task. The essay may not take a position on the issue, or the essay may take a position but fail to convey reasons to support that position, or the essay may take a position but fail to maintain a stance. There is little or no recognition of a counterargument to the writer's position. The essay is thinly developed. If examples are given, they are general and may not be clearly relevant. The essay may include extensive repetition of the writer's ideas or of ideas in the prompt. Focus on the general topic is maintained, but focus on the specific issue in the prompt may not be maintained. There is some indication of an organizational structure, and some logical grouping of ideas within parts of the essay is apparent. Transitions, if used, are simple and obvious, and they may be inappropriate or misleading. An introduction and conclusion are discernible but minimal. Sentence structure and word choice are usually simple. Errors may be frequently distracting and may sometimes impede understanding.

Score = 1

Essays within this score range show little or no skill in responding to the task.

The essay shows little or no understanding of the task. If the essay takes a position, it fails to convey reasons to support that position. The essay is minimally developed. The essay may include excessive repetition of the writer's ideas or of ideas in the prompt. Focus on the general topic is usually maintained, but focus on the specific issue in the prompt may not be maintained. There is little or no evidence of an organizational structure or of the logical grouping of ideas. Transitions are rarely used. If present, an introduction and conclusion are minimal. Sentence structure and word choice are simple. Errors may be frequently distracting and may significantly impede understanding.

No Score

Blank, Off-Topic, Illegible, Not in English, or Void

How to Score the Writing Test

Two trained readers will score your essay on the actual Writing Test. These readers are trained by reading examples of papers at each score point and by scoring many practice papers. They are given detailed feedback on the correctness of their scores during practice. During actual scoring, score differences of more than one point will be evaluated by a third trained reader to resolve discrepancies. This method is designed to be as objective and impartial as possible. So—how can you rate your *own* practice Writing Test?

It is difficult to be objective about one's own work, and you have not had the extensive training provided to actual readers of the ACT Writing Test. However, it is to your advantage to read your own writing critically. Becoming your own editor helps you grow as a writer and as a reader. So it makes sense for you to evaluate your own practice essay. It may also be helpful for you to give your practice essay to another reader to get another perspective: perhaps that of a classmate, a parent, or an English teacher, for example. Thinking and talking with others about writing is good preparation for the ACT Writing Test. To rate your essay, you and your reader(s) should read the scoring guidelines and example essays, which begin below and continue through page 71, and then assign your practice essay a score of 1 through 6.

For an actual administration, each essay will be scored on a scale from 1 (low) through 6 (high). The score is based on the overall impression that is created by all the elements of the writing. The scores given by the two readers are added together, yielding the Writing subscore range 2–12 shown in Table 4 on page 72.

Scoring Guidelines (see page 64)

These are the guidelines that will be used to score your essay. These guidelines are also called a “rubric.” Many papers do not fit the exact description at each score point. You should note that the rubric says: “Papers at each level exhibit *all* or *most* of the characteristics in the descriptors.” To score your paper, read your response and try to determine which score point and paragraph in the rubric best describes most of the characteristics of your essay.

Then (because your Writing Test subscore is the sum of two readers' ratings of your essay), you should multiply your 1–6 score by 2 when you use Table 4, on page 72, to find your Combined English/Writing score. Or, if both you and someone else read and score your practice essay, add those scores together.

Comparing Your Scores

The Writing Test norms table (Table 3B on page 63) allows you to compare your score on the practice Writing Test with the scores of recent high school graduates who took the ACT Plus Writing. The norms for the Writing Test are reported the same way as the norms for the multiple-choice tests (see page 57). For example, a Writing subscore of 8 has a cumulative percent of 86. This means that 86% of students had a Writing subscore of 8 or lower. Remember that your scores and percents at or below are only *estimates* of the scores you will obtain on an actual administration of the ACT Plus Writing. They should be considered in connection with your performance on other essay tests and your planned college curriculum.

College Readiness Standards

The College Readiness Standards for Writing (see page 57) can be found at www.act.org/standard.

Example Essays and Scoring Explanations

Readers for the ACT Writing Test are trained by scoring many essays before they score “live” essays. Although we cannot provide you with the same extensive training these readers receive, reading the example essays that follow will help you better understand some of the characteristics of essays at each score point. You will also be able to read a brief explanation of how each essay was scored. The example essays are in response to the practice prompt on page 56.

Score = 1

It is not to distacting for students because everybody has there own way of doing there homework. If a kid wants to watch tv while he is doing his homework then let him. That is his comfort zone that is when most kids are relaxed. Thats why they finish there homework. It also makes them calm so there not all worried if there almost done or not because there doing there work but at the same time there also relaxed because there watching the favorite cartoon or reaility show or whatever they watch. But you can't let them watch the tv more than there doing there homework because then nothing will get done like that.

Score Point 1 Scoring Explanation

Essays that earn a 1 show little or no skill in responding to the task. This essay begins with the writer's position statement, but because no context is provided for the discussion, it is difficult to discern the exact question the student is responding to. In the second sentence, the writer makes the argument a little clearer by saying that *If a kid wants to watch tv while he is doing his homework then let him*, but nowhere in the essay does the writer refer to the prompt's specific question about multitasking.

The essay is minimally developed. Rather than offering specific ideas to support the writer's position, the essay merely repeats the general assertion that watching television helps a student relax (*That is his comfort zone that is when most kids are relaxed.... It also makes them calm so there not all worried...but at the same time there also relaxed*).

There is no evidence of an organizational structure. The writer does not offer an introduction, and instead of providing a conclusion, the writer ends the essay with only the clarification that *you can't let them watch the tv more than there doing there homework because then nothing will get done like that*. In addition, the writer uses only a single transition to connect ideas (*also*).

Errors in sentence structure, such as run-on sentences, and frequent misspellings (*there* used instead of *their* and *they're*) are distracting but do not impede understanding.

Score = 2

Multitasking is a good practice while doing homework because a lot of careers need good multitaskers or student might have less time to do things. Multitasking is a necessity in today modern lifestyle. You need to be able to keep track of multiple things like the family, taxes, arrangements, meetings....etc., the list goes on. To develop multitasking at a young age is critical. What if you need to do 10 things in a day or have to get the kids ready for school in the morning, while you're on an important phone call with your boss? If you became a business owner, you have to keep up with supply + demand of products and services, keep up with customer satisfaction. These are all instances where multitasking is irreplaceable. Multitaskers are always needed for jobs. Some educators say multitasking reduces grades. I say different being a multitasker + getting a 3.5 GPA. I watch a movie while typing a paper and it sometimes helps clear writer's block and allows the ideas to flow. Take it from a multitasker we're always needed.

Score Point 2
Scoring Explanation

Essays that earn a 2 demonstrate inconsistent or weak skill in responding to the task. This writer takes the position that multitasking is a good practice, arguing that *a lot of careers need good multitaskers* and that without multitasking *student might have less time to do things*. There is a slight recognition of complexity at the end of the essay as the writer briefly addresses a counter argument (*Some educators say multitasking reduces grades....*), but the writer offers only a couple of assertions about his or her personal experience to refute the point.

Development of ideas is thin. The writer offers examples of situations when multitasking might be necessary (being on the phone while trying to get kids to school, keeping up with supply and demand and customer satisfaction as a business owner), but development of these ideas is limited to single-sentence assertions. In addition, beyond the brief personal example of watching a movie while typing a paper, the writer provides no discussion of the point that students will have less time to do things if they don't multitask.

There is little indication of an organizational structure. Ideas are presented mostly as a random list, with only occasional evidence of the writer having attempted to group related points logically. Transitions are not used to connect ideas, and there is no clear evidence of an introduction or conclusion, beyond the opening position statement and the closing declaration, *Take it from a multitasker we're always needed*.

Sentence structure and word choice are simple, and when the writer attempts more complex sentence structures, errors such as run-on sentences and fragments result (*I say different being a multitasker + getting a 3.5 GPA.*). These errors and frequent spelling mistakes (*practice, necessity, develop, age, customer, irreplaceable*) are distracting but do not impede understanding.

Score = 3

Multitasking is a very bad habit when doing homework. It has negative effects on concentration, learning skills and the homework itself. By eliminating multitasking during homework would have a positive influence on many aspects.

Multitasking while doing homework affects concentration. While doing homework, a student should concentrate only on the homework. If the student is doing another task along with the homework, they are not as concentrated on the homework. Less concentration on the homework will lead to the student not doing as well as they would have if they had done only the homework.

Multitasking prevents students from doing their very best. It is a poor learning habit that can be harmful to the student in the long run. Multitasking prevents students from learning all that they can. When asked to recall something from an assignment, a student will not be able to recall everything they should because they were multitasking while doing an assignment.

The homework itself is influenced negatively by multitasking. The student may receive a poor grade because they did not devote their full attention to the homework. Multitasking will not let the very best a student can do shine through. It will prevent good grades and proper learning.

Multitasking certainly does not help student in their concentration, learning skills or their homework. In the future, we will likely see the long-term effects that multitasking has had on people's learning.

Score Point 3
Scoring Explanation

Essays that earn a 3 demonstrate some developing skill in responding to the task. This essay does not provide any context for the discussion, and instead begins with the writer's position and then offers a list of three reasons why multitasking is a *very bad habit* for students.

Development of the writer's three main ideas is limited and repetitive—each body paragraph consists mostly of a series of assertions rather than movement from general ideas to specific reasons or examples. For instance, as the writer tries to argue the first idea that multitasking negatively affects students' ability to concentrate, the writer provides only the unsupported assertions that *While doing homework, a student should concentrate only on the homework* and that *If the student is doing another task along with the homework, they are not as concentrated on the homework*. Further, all three body paragraphs argue essentially the same point—that multitasking *prevents students from doing their very best*.

The essay's organization is simple; the structure of the essay follows the order of the three points laid out in the introduction. Ideas are logically grouped, in that sentences in each paragraph contain related ideas, but there is little evidence of logical sequencing of ideas within paragraphs and there are no transitions linking ideas between one paragraph and the next. The introduction consists of little more than the position statement, and the conclusion merely paraphrases the introduction.

Language demonstrates basic control, although sentence structures show little variety (*Multitasking prevents students from doing...Multitasking prevents students from learning...*). Word choice, however, is clear, if somewhat repetitive. There are no distracting language errors.

Score = 4

Many people say that multitasking while doing homework is a bad practice because it negatively affects the quality of a student's work. Others, however, think that multitasking while doing homework is good because students accomplish more during their limited free time. I, on the otherhand, believe that multitasking while doing homework is both good and bad, depending on the other activities a student partakes in while completing his or her homework.

Multitasking while doing homework can be a bad practice when a student is not doing something productive. For example, watching videos on YouTube or reality-like shows while doing homework would be a bad practice because a student will tend to be more engaged in what he/she is watching rather than the homework that is due the following day. Surfing the internet or logging into Facebook while trying to complete homework is also bad. Not only do these action distract students from finishing their homework, but it also causes them to spend more time trying to complete his or her homework. When a student has too many distractions, it'll usually take more time for him/her to finish homework.

Although multitasking while doing homework is sometimes a bad practice, it can also be a good thing. Multitasking while doing homework can be a good thing when a student is doing another thing that is productive, unlike sitting in front of a TV with a bag of chips and at the same time doing homework. If a student is on the internet doing research for another class while doing homework, then multitasking is ok because the student is doing something for his/her class. By multitasking productively, students learn how to manage their time. Students become better prepared for the real world, where adults tend to multitask almost everyday.

For me, multitasking while doing homework is hard. Sometimes I tend to watch T.V. while trying to complete my homework, and I end up spending more time trying to finish my homework. If I'm doing something unproductive while doing my homework, I become too distracted and the quality of my homework is not as good as it should be. At other times, however, doing something productive while doing homework allows me to finish my work earlier than I usually do. With extra free time, I am able to study for upcoming tests, or quizzes, and even go outside and enjoy a bright and sunny day.

Depending on how a person looks at multitasking while doing homework, it can be a good or bad practice. As for me, multitasking can be good or bad depending on the other activities you're involved in while trying to complete your homework.

Score Point 4 Scoring Explanation

Essays that earn a 4 demonstrate adequate skill in responding to the task. This writer provides some context for the discussion by restating the two opposing sides of the issue presented in the prompt, and takes a compromise position on the debate, arguing that whether multitasking while doing homework is good or bad depends on what other activities the student is doing at the same time. The writer demonstrates some recognition of complexity by mentioning some long-term implications of multitasking (*By multitasking productively, students learn how to manage their time. Students become better prepared for the real world, where adults tend to multitask almost everyday.*).

Development of the writer's ideas is adequate, with some movement from general ideas to specific examples. For example, in the first body paragraph, the writer begins by saying *Multitasking while doing homework can be a bad practice when a student is not doing something productive*, and then goes on to elaborate with examples of unproductive activities that can't be done while doing homework (watching videos on YouTube, watching reality TV, surfing the Internet). The writer then also provides supporting explanations for why such activities can't be combined with homework (*a student will tend to be more engaged in what he/she is watching rather than the homework that is due the following day*).

The organization of the essay is clear. There is some evidence of the logical sequencing of ideas within paragraphs. In addition, the writer transitions smoothly from the negative effects of multitasking to the positive effects by using an integrated transition between paragraphs (*Although multitasking while doing homework is sometimes a bad practice, it can also be a good thing.*). Other, simpler transitions are also used to show the connection of ideas (*on the otherhand, however, also*). Although the introduction is somewhat developed and includes some context, the conclusion is underdeveloped and offers only a brief restatement of the writer's position.

Language control is adequate. There is some sentence variety, and despite some general and repetitive phrasing (*Multitasking while doing homework can be a bad practice Multitasking while doing homework can be a good thing*), word choice is mostly appropriate (*limited free time, partakes, engaged, surfing the internet*). Language errors are not distracting and do not impede understanding.

Score = 5

Multitasking has become an essential talent for high school students nowadays. In the face of sports, family, homework, friends, and all the other obligations and activities that such students often have to deal with, time management is essential to maintaining the delicate balance between sanity and simply being overwhelmed. As well, there are many positive effects (both short-term and long-term) that can result from doing several things at once. Multitasking is very useful for managing one's time and in gaining beneficial habits.

Time-management has become quite a difficult problem for high school students today. With all the pressures of taking many substantive classes, especially ones with potential college-credit, the average student workload has increased drastically over the years. Added to the homework and studying resulting from such classes, students participate in extracurricular activities varying from clubs and sports teams to religious organizations and volunteer work. Thus, while the day has stayed constant at 24 hours, the amount of activity within that time frame has increased. There are several options for students in such situations. They can go the route of sleep deprivation (endangering their health), they can try to do things individually (resulting in a decline in quality of work or the compulsory dropping of several obligations), or they can manage their time so as to do more than one task at once.

If the student does choose to multi-task, there can actually be benefits in the form of good habits. In the short-term, the student may actually create enough of a gap in his or her schedule to allow for some leisure time, providing an often much needed break from work. The long-term effects are even more beneficial in that the student will form useful talents that will serve them well in the business world. In today's fast-paced society, the ability to multi-task is highly valued and one who is greatly practiced in it will be more likely to succeed than one who isn't. Some people may claim that it isn't possible to divide attention evenly between several things at once while producing something of optimum quality. This view, though, is only correct when it comes to multitasking many things. As long as there are not too many simultaneous tasks, the quality can still be high.

In short, multitasking is a very valuable tool for high school students and one which they should all utilize. It allows for healthy and efficient time management in a time of pressure and high expectations. It can also cause the formation of useful talents for later in life. Multitasking, therefore, should be accepted, and even encouraged by teachers and parents as a tool for high school students.

Score Point 5
Scoring Explanation

Essays that earn a 5 demonstrate competent skill in responding to the task. This writer begins with a somewhat broad context that describes the value of being able to multi-task in order to maintain a *delicate balance* among all the *obligations and activities* students must frequently handle. The writer takes a position in favor of multitasking, arguing that *Multitasking is very useful for managing one's time and in gaining beneficial habits*.

The writer demonstrates recognition of complexity by briefly responding to a counter argument (*Some people may claim that it isn't possible to divide attention evenly between several things at once while producing something of optimum quality. This view, though, is only correct when it comes to multitasking many things. As long as there are not too many simultaneous tasks, the quality can still be high.*). The writer also demonstrates a recognition of complexity by partially evaluating some long-term implications of his or her position, arguing that learning how to multitask will help students form long-term habits that will serve them in the future (*In today's fast-paced society, the ability to multi-task is highly valued and one who is greatly practiced in it will be more likely to succeed than one who isn't.*).

Development of ideas is specific and logical, with clear movement between general statements and specific reasons and examples. For example, the writer begins the first body paragraph by describing the increased pressure on students to take more substantive classes and participate in a variety of extracurricular activities (clubs, sports teams, religious organizations, volunteer work). The writer then goes on to argue that, although there might be other options for students to employ when dealing with these increased pressures, alternatives to multitasking have negative consequences that multitasking does not have, such as sleep deprivation and declining quality of work.

Organization of the essay is clear. Ideas are logically sequenced—each sentence leads logically to the next. The writer also uses both simple (*As well,...*) and integrated transitions to connect ideas (*Added to the homework and studying resulting from such classes, students participate in extra curricular activities....; This view, though, is only correct when...*). The introduction and conclusion are both clear and generally well developed. The introduction provides context for the prompt's issue, and the conclusion summarizes the essay's main points.

Language is competent. Sentence structures are varied and word choice is varied and sometimes precise (*essential talent, compulsory, optimum quality*).

Score = 6

History research paper. Organic Chemistry lab write-up. To Kill a Mockingbird reading. Although Twittering, texting, or surfing the Internet may very well hold infinitely more appeal to high schoolers, how can all these unrelated activities not affect one's schoolwork? As our world becomes more and more technologically superior, teens are in their element on the high-def, fast-paced scene that is modern American life today. Sure it's fun, current, and hip, but when youngsters focus more on social sites and techno activities, what are they really sacrificing?

While it seems that taking periodic breaks from study sessions and school assignments to partake in a loved activity keeps students more focused and able to finish homework at night, multitasking is not the same action. What would have been a "quick fifteen-point study guide" could very likely turn into a three-hour-long, TV-dazed catastrophe because the season finale of American Idol just flashed onto the TV screen. A much more effective—and rewarding—plan of attack would be to delegate homework commitments ahead of a special event so that students can participate in a long-awaited activity free from the burden of a looming assignment.

Likely enough, the quality of work students effect will be much higher and richly insightful if a block of time can be stowed away for disciplined use. It seems to be common sense that a persuasive essay on Cold War policies would be crafted much more masterfully if a student's brain wasn't bouncing between Stalin, a best friend's date crisis, SALT I Accords, Mom yelling to set the dinner table, and presidential influences in the 1960s. Slow down! So much information that is so distantly related to each other cannot possibly combine to earn a hardworking student an "A" on an assignment. The time "saved" by "accomplishing" several tasks at once—if there even is a reduction in time—is most definitely wiped out after a sullen-faced teenager is forced to bring home a report card that represents negligent schoolwork.

Moreover, multitasking is a bad habit that will be even more difficult to break as one matriculates through college and budding professionalism. On the job, attempting to finish two, three, or more jobs on a timely basis and with high quality, no less, will be an exercise in futility. If an employee couldn't find the time to complete such tasks before the crunch, what's to say they will be successful now that they have three vital duties to execute? Good luck with that promotion.

The bottom line on multitasking is that today's teenagers are already overwhelmed in a world where technology and communications experiences exponential growth each year. With a myriad of choices in front of them, it is their duty to their futures that they finish essential tasks—such as homework—before enjoying other such pleasures and pleasantries. In a world where quantity is exploding and standards for quality rise with each passing year, humanity as a whole has been forced to quickly rise to the occasion. So, how much do you want success? Twitter and Facebook will always be there, but will that Presidential Scholarship to your first-choice Ivy League school wait indefinitely while you forward one more e-mail?

Score Point 6 Scoring Explanation

Essays that earn a 6 demonstrate effective skill in responding to the task. This essay opens with a broad context that juxtaposes examples of the variety of assignments teens must complete with the technological distractions facing teens. The writer then critically describes the increasingly *hi-def, fast-paced scene that is modern American life* and acknowledges that although this world is fun and hip, when teens focus too heavily on these distractions, something is sacrificed.

The essay demonstrates complexity by exploring the long-term implications of multitasking, arguing that it is *a bad habit that will be even more difficult to break as one matriculates through college and budding professionalism*. The writer suggests that by trying to juggle too many activities at once, students will do poorly in all of them, and this trend will have more negative consequences as students move into adulthood. The writer also addresses some complications of the issue by questioning whether multitasking even leads to the kind of time-saving that its proponents claim it does (*The time "saved" by "accomplishing" several tasks at once—if there even is a reduction in time—is most definitely wiped out after a sullen-faced teenager is forced to bring home a report card that represents negligent schoolwork.*).

While the writer's ideas are not developed evenly across all the paragraphs, development overall is ample and persuasive. The writer elaborates upon general statements (*the quality of work students effect will be much higher and richly insightful if a block of time can be stowed away*) by supporting them with specific details and hypothetical situations (*a persuasive essay on Cold War policies would be crafted much more masterfully if a student's brain wasn't bouncing between Stalin, a best friend's date crisis, SALT I Accords, Mom yelling to set the dinner table, and presidential influences in the 1960s*).

The essay's organization is clear and grows organically from the writer's purpose: although the writer does not explicitly take a position in the introduction, the rhetorical question that ends the paragraph implies the writer's position, which is then elaborated on over the course of the essay. Ideas are logically sequenced. While some of the transitions between paragraphs are somewhat predictable (*Moreover, The bottom line*), others are smoothly integrated, as in the transition from the introduction into the first body paragraph (*While it seems that taking periodic breaks from study sessions and school assignments ... keeps students more focused and able to finish homework at night, multitasking is not the same action.*). Both the introduction and conclusion are effective, clear, and well developed, as the writer skillfully weaves in specific details and rhetorical questions (*So, how much do you want success? Twitter and Facebook will always be there, but will that Presidential Scholarship to your first-choice Ivy League school wait indefinitely while you forward one more e-mail?*) while presenting and reiterating the essay's main ideas.

The essay shows a good command of language. The essay's varied sentence structures and precise word choice (*sullen-faced teenager, budding professionalism, exercise in futility*) allow the writer to maintain a critical tone throughout. There are few errors in this essay, and the occasional usage errors (*richly insightful*) are too minor to be distracting.

TABLE 4**Calculating Your Combined English/Writing Score**

Complete these steps to calculate your Combined English/Writing score for your practice tests.

1. Locate your scale score for the English Test on page 61 and enter it here: _____.
2. Enter your Writing Test score (1–6) here _____ and double it to get your Writing subscore (2–12): _____
(If two people read and scored your Writing Test, add those two scores to get your Writing subscore.)
3. Use the table below to find your Combined English/Writing score.

- First, circle your ACT English Test score in the left column.
- Second, circle your ACT Writing subscore at the top of the table.

- Finally, follow the English Test score row across and the Writing subscore column down until the two meet. Circle the Combined English/Writing score where the row and column meet. (For example, for an English Test score of 19 and a Writing subscore of 6, the Combined English/Writing score is 18.)

4. Using the number you circled in the table below, write your Combined English/Writing score here: _____.
(The highest possible Combined English/Writing score is 36 and the lowest possible score is 1.)

ACT English Test score _____

Writing subscore _____

Combined English/Writing Score _____
(from table below)

Combined English/Writing Scale Scores											
English Test Score	Writing Subscore										
	2	3	4	5	6	7	8	9	10	11	12
1	1	2	3	4	5	6	7	8	9	10	11
2	2	3	4	5	6	6	7	8	9	10	11
3	2	3	4	5	6	7	8	9	10	11	12
4	3	4	5	6	7	8	9	10	11	12	13
5	4	5	6	7	8	9	10	11	12	13	14
6	5	6	7	7	8	9	10	11	12	13	14
7	5	6	7	8	9	10	11	12	13	14	15
8	6	7	8	9	10	11	12	13	14	15	16
9	7	8	9	10	11	12	13	13	14	15	16
10	8	9	9	10	11	12	13	14	15	16	17
11	8	9	10	11	12	13	14	15	16	17	18
12	9	10	11	12	13	14	15	16	17	18	19
13	10	11	12	13	14	14	15	16	17	18	19
14	10	11	12	13	14	15	16	17	18	19	20
15	11	12	13	14	15	16	17	18	19	20	21
16	12	13	14	15	16	17	18	19	20	20	21
17	13	14	15	16	16	17	18	19	20	21	22
18	13	14	15	16	17	18	19	20	21	22	23
19	14	15	16	17	18	19	20	21	22	23	24
20	15	16	17	18	19	20	21	21	22	23	24
21	16	17	17	18	19	20	21	22	23	24	25
22	16	17	18	19	20	21	22	23	24	25	26
23	17	18	19	20	21	22	23	24	25	26	27
24	18	19	20	21	22	23	23	24	25	26	27
25	18	19	20	21	22	23	24	25	26	27	28
26	19	20	21	22	23	24	25	26	27	28	29
27	20	21	22	23	24	25	26	27	28	28	29
28	21	22	23	24	24	25	26	27	28	29	30
29	21	22	23	24	25	26	27	28	29	30	31
30	22	23	24	25	26	27	28	29	30	31	32
31	23	24	25	26	27	28	29	30	30	31	32
32	24	25	25	26	27	28	29	30	31	32	33
33	24	25	26	27	28	29	30	31	32	33	34
34	25	26	27	28	29	30	31	32	33	34	35
35	26	27	28	29	30	31	31	32	33	34	35
36	26	27	28	29	30	31	32	33	34	35	36

Marking Directions: Mark only **one** oval for each question. Fill in response completely. Erase errors cleanly without smudging.

Correct mark: ☐ ☒ ☐ ☐

Do NOT use these incorrect or bad marks.

Incorrect marks: ☒ ☒ ☒ ☒
 Overlapping mark: ☐ ☐ ☒ ☒
 Cross-out mark: ☐ ☒ ☐ ☐
 Smudged erasure: ☐ ☐ ☐ ☐
 Mark is too light: ☐ ☐ ☐ ☐

BOOKLET NUMBER

1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9
0	0	0	0	0	0

FORM

--	--	--

BE SURE TO FILL IN THE CORRECT FORM OVAL.

○ 67C

Print your 3-character **Test Form** in the boxes above and fill in the corresponding oval at the right.

TEST 1

1 (A B C D)	14 (F G H J)	27 (A B C D)	40 (F G H J)	53 (A B C D)	66 (F G H J)
2 (F G H J)	15 (A B C D)	28 (F G H J)	41 (A B C D)	54 (F G H J)	67 (A B C D)
3 (A B C D)	16 (F G H J)	29 (A B C D)	42 (F G H J)	55 (A B C D)	68 (F G H J)
4 (F G H J)	17 (A B C D)	30 (F G H J)	43 (A B C D)	56 (F G H J)	69 (A B C D)
5 (A B C D)	18 (F G H J)	31 (A B C D)	44 (F G H J)	57 (A B C D)	70 (F G H J)
6 (F G H J)	19 (A B C D)	32 (F G H J)	45 (A B C D)	58 (F G H J)	71 (A B C D)
7 (A B C D)	20 (F G H J)	33 (A B C D)	46 (F G H J)	59 (A B C D)	72 (F G H J)
8 (F G H J)	21 (A B C D)	34 (F G H J)	47 (A B C D)	60 (F G H J)	73 (A B C D)
9 (A B C D)	22 (F G H J)	35 (A B C D)	48 (F G H J)	61 (A B C D)	74 (F G H J)
10 (F G H J)	23 (A B C D)	36 (F G H J)	49 (A B C D)	62 (F G H J)	75 (A B C D)
11 (A B C D)	24 (F G H J)	37 (A B C D)	50 (F G H J)	63 (A B C D)	
12 (F G H J)	25 (A B C D)	38 (F G H J)	51 (A B C D)	64 (F G H J)	
13 (A B C D)	26 (F G H J)	39 (A B C D)	52 (F G H J)	65 (A B C D)	

TEST 2

1 (A B C D E)	11 (A B C D E)	21 (A B C D E)	31 (A B C D E)	41 (A B C D E)	51 (A B C D E)
2 (F G H J K)	12 (F G H J K)	22 (F G H J K)	32 (F G H J K)	42 (F G H J K)	52 (F G H J K)
3 (A B C D E)	13 (A B C D E)	23 (A B C D E)	33 (A B C D E)	43 (A B C D E)	53 (A B C D E)
4 (F G H J K)	14 (F G H J K)	24 (F G H J K)	34 (F G H J K)	44 (F G H J K)	54 (F G H J K)
5 (A B C D E)	15 (A B C D E)	25 (A B C D E)	35 (A B C D E)	45 (A B C D E)	55 (A B C D E)
6 (F G H J K)	16 (F G H J K)	26 (F G H J K)	36 (F G H J K)	46 (F G H J K)	56 (F G H J K)
7 (A B C D E)	17 (A B C D E)	27 (A B C D E)	37 (A B C D E)	47 (A B C D E)	57 (A B C D E)
8 (F G H J K)	18 (F G H J K)	28 (F G H J K)	38 (F G H J K)	48 (F G H J K)	58 (F G H J K)
9 (A B C D E)	19 (A B C D E)	29 (A B C D E)	39 (A B C D E)	49 (A B C D E)	59 (A B C D E)
10 (F G H J K)	20 (F G H J K)	30 (F G H J K)	40 (F G H J K)	50 (F G H J K)	60 (F G H J K)

TEST 3

1 (A B C D)	8 (F G H J)	15 (A B C D)	22 (F G H J)	29 (A B C D)	36 (F G H J)
2 (F G H J)	9 (A B C D)	16 (F G H J)	23 (A B C D)	30 (F G H J)	37 (A B C D)
3 (A B C D)	10 (F G H J)	17 (A B C D)	24 (F G H J)	31 (A B C D)	38 (F G H J)
4 (F G H J)	11 (A B C D)	18 (F G H J)	25 (A B C D)	32 (F G H J)	39 (A B C D)
5 (A B C D)	12 (F G H J)	19 (A B C D)	26 (F G H J)	33 (A B C D)	40 (F G H J)
6 (F G H J)	13 (A B C D)	20 (F G H J)	27 (A B C D)	34 (F G H J)	
7 (A B C D)	14 (F G H J)	21 (A B C D)	28 (F G H J)	35 (A B C D)	

TEST 4

1 (A B C D)	8 (F G H J)	15 (A B C D)	22 (F G H J)	29 (A B C D)	36 (F G H J)
2 (F G H J)	9 (A B C D)	16 (F G H J)	23 (A B C D)	30 (F G H J)	37 (A B C D)
3 (A B C D)	10 (F G H J)	17 (A B C D)	24 (F G H J)	31 (A B C D)	38 (F G H J)
4 (F G H J)	11 (A B C D)	18 (F G H J)	25 (A B C D)	32 (F G H J)	39 (A B C D)
5 (A B C D)	12 (F G H J)	19 (A B C D)	26 (F G H J)	33 (A B C D)	40 (F G H J)
6 (F G H J)	13 (A B C D)	20 (F G H J)	27 (A B C D)	34 (F G H J)	
7 (A B C D)	14 (F G H J)	21 (A B C D)	28 (F G H J)	35 (A B C D)	

ACT STUDENT REVIEW: The test administrator will give you instructions for completing this section.



Student Review: Your responses to these items will assist ACT and your test center in providing the best possible conditions for testing and planning for the future. Fill in the oval indicating your response to each item printed on the back of your test booklet.

Yes	No	Yes	No	Yes	No
1 ○	○	6 ○	○	11 ○	○
2 ○	○	7 ○	○	12 ○	○
3 ○	○	8 ○	○	13 ○	○
4 ○	○	9 ○	○	14 ○	○
5 ○	○	10 ○	○	15 ○	○

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✧ **ACT English Practice Tests:**

<https://www.crackab.com/act/english/>

✧ **ACT Math Practice Tests:**

<https://www.crackab.com/act/math/>

✧ **ACT Reading Practice Tests:**

<https://www.crackab.com/act/reading/>

✧ **ACT Science Practice Tests:**

<https://www.crackab.com/act/science/>

ACT Grammar: <https://www.crackab.com/act/grammar/>

ACT Real Past Papers Download:

<https://www.crackab.com/act-downloads/>

Digital SAT & New SAT Practice Tests:

<https://www.cracksat.net>

Real SAT Tests Download:

<http://www.cracksat.net/sat-downloads/>

AP Exams Practice Tests:

<https://www.crackap.com>

<https://www.apstudy.net>

Use a soft lead No. 2 pencil only. Do NOT use a mechanical pencil, ink, ballpoint, or felt-tip pen.

Print your 6-digit
Booklet Number
in the boxes at the
right.

--	--	--	--	--	--

--	--	--

Print your
3-character
Test Form in
the boxes above
and fill in the
corresponding
oval at the right.

○ 14R

Begin WRITING TEST here.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

1

IM-188616-001:654321

www.actexam.net

[illegible]

Do not write in this shaded area.

Cut Here

WRITING TEST

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

3

If you need more space, please continue on the next page.

PLEASE DO NOT WRITE IN THIS AREA.

[illegible]

[illegible]

4

Do not write in this shaded area.